GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

QIAN QI

Abstract. We study finite-resource prediction in positive prepared causal experiments using an actually prepared comparison law that need not belong to the operating model family. A reference-side converse replaces only each productive report–audit row while retaining the actual pre-call occupation. The resulting positive submeasure has controlled missing mass, a cost-certified density cap and controlled task moments, giving a lower bound uniform over all legal strategies. A common parameter-independent approximate balancing program gives a matching upper at the same expected physical budget. The bounds recover exact stopped risk profiles at zero error and matching acquired-geometry orders on a quantified two-sided calibration regime. Drift certificates extend the converse to metastable and singular calibration with explicit preparation mass and residence time. Classical controlled channels and noncommuting quantum instruments verify the same theorem without an exactly fair operating reference or a posterior-rank floor. A same-task extension combines attained calibration and causal erasure error. Event-stopped observers may reuse live labels for outputs, whereas intrinsically state-stopped observers pay for disjoint live and terminal roles; both resource laws are proved. We retain nonintervening Gaussian sensing with a joint waiting–output geometry. Numerical precision, program description, workspace, physical time and simulator states are accounted for separately.

1. Introduction

A prepared causal experiment determines which future tests can be predicted from its reports. The predictive quotient records this equivalence of executable tests, but does not by itself determine the geometry that a finite observer can acquire. Waiting for a rare informative report changes both the attainable law and the physical cost. Persistent memory constrains the route to that report as well as its final description. A further difficulty is calibration: a comparison law may describe a physically prepared ideal experiment accurately without being an exact member of the operating model family. A converse that silently treats it as an admitted model is then unjustified.

We establish a two-sided resource–risk transfer under explicit raw experiment certificates. One actual witness bounds the effort of every competing strategy and the accumulated discrepancy of its productive report–audit rows from a prepared comparison law. Separately, one parameter-independent acquisition program approximately balances the report and task measures in every operating model. Our center is Theorem 4.1. It uses neither an exact comparison model in the uncertainty family nor a comparison execution that stops under the competing program.

The main construction in the converse is an accepted comparison submeasure. Retain the actual distribution of each pre-call history, but replace its current productive report–audit row only. The resulting positive measure has a controlled mass defect and a density cap set by actual expected acquisition count. It need not be an executable terminal law. Its unnormalized task moments nevertheless give an all-strategy risk lower. This separates a valid local comparison from the generally invalid replacement of an entire selected path law.

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Write $\mathcal{Q}_m(b)$ for the marked trimmed-quantization profile derived from the comparison's report law ν, audit mean f, and residual variance w; its cap b retains actual preparation mass. In the matched-effort, identity-export case the central bounds take the form

$$[t\mathcal{Q}_m(b/t) - \mathfrak{e}_N]_+ \ \le R \ \le (1+ba_+)\mathcal{Q}_m\Big(\frac{b}{1+ba_+}\Big) + be_+, \qquad t = (1-\mathfrak{a}_N)_+.$$

Here the lower is read as zero at $t = 0$, and the upper has its stated feasibility threshold. The lower defect bounds are actual accumulated charges, not a best-policy occupation estimate. Different effort constants and export multiplicities remain explicitly distinct in the theorem. The formula recovers exact equalities at zero errors and matching geometric orders on a quantitative two-sided calibration regime.

The output width depends on the physical stopping convention. A state-stopped observer has a state-only halt map; its live and terminal labels are disjoint, giving width $M - B$ when B live labels are mandatory and sufficient. An event-stopped observer emits its final commitment during a transition and may reuse live labels as final outputs, giving width M. We retain both exact resource laws and their proof, rather than excluding reuse silently. All intercall phases, clocks and continuing random seeds are charged.

Theorem 5.2 replaces small uniform defects by a drift certificate on actual good/bad mode occupation. The initial bad mass and its residence time appear explicitly; singular and changing-rank report kernels are permitted. Theorems 6.1 and 6.2 verify the same central theorem from a classical controlled sign channel and a noncommuting positive quantum instrument. Their nominal balanced preparations need not be operating models. A joint calibration–erasure extension keeps all errors in the same terminal squared loss, while finite numerical execution and causal simulator costs are stated separately.

We also retain a different, nonintervening raw application. A current hidden unit vector is observed through Gaussian noise and is not altered by the observer. Theorem 16.1 proves a matching law

$$(1+\log b)^{-1/2} + m^{-2/(d-1)}$$

against every legal stopping rule, together with exact finite-output obstructions and a sharp noise constant. This application is not mislabeled as a post-command balanced task, and its physical posterior is derived directly from the Gaussian kernel.

The variational profile itself has an established ancestry. When $w = 0$, writing $a = 1/b$ and $g = h/b$ gives $0 \le g \le 1$, $\int g\, d\nu = a$, and the objective is the normalized distortion of a fractionally retained mass a. This is the population trimming formulation underlying trimmed k-means and impartial trimming [13], [14]. With $w \neq 0$ it is the corresponding marked objective. We do not claim the density cap or low-distortion selection principle as new. Classical reject-option prediction [5], [6], invariant decision procedures [3], and quantum randomization [4] supply related tools. Approximate rejection sampling from a fixed independent batch [15], [16] has a different budget and output objective: here physical attempts can be history dependent, the criterion is an audited decision loss, and a single finite-state program must work for every admitted model.

Almost-sure expected-cost reachability in partially observed control is studied in [11], and fixed-memory limit-sure reachability in [12]. Sufficiency results for chattering policies in absorbing decision processes are developed in [10]. These works concern policy classes and reachability or occupation criteria; the present matching assertions require, in addition, the specified report–audit balance and acquisition certificates.

Accumulation of local error along transient occupation and Lyapunov telescoping are standard stochastic-control techniques. In particular, the occupation-weighted rollout bounds of Hansson and Wahlberg [17] address policy-specific Bellman approximation and properness. We do not claim occupation weighting, drift control, or the mere appearance of an expected-time factor as new. Here the comparison submeasure controls both its lost mass and its task moment against a trimmed

finite-dictionary optimum, yielding a converse uniform over all admissible observers even when the comparison process is not an admitted or terminating execution. This combines a different quantifier structure, a selected geometric objective and explicit stopping-resource widths.

The result is a sufficient-class theorem, not a classification of all positive kernels. Measurable quotient realization has its separate hypotheses in Appendix A; arbitrary projective arrays are not declared physical states. Our proofs distinguish new comparison arguments from inherited exact balance and role counting, and from standard conditional projection, compactness, trimming, Gaussian tails and covering. Ideal Borel existence is not finite algorithmic synthesis. The certificates needed to learn unknown kernels, to price every computation optimally, or to treat unbounded losses and nonterminating protocols are not supplied by the present theorem.

2. Prepared experiments and stopping resources

All state, action and report spaces are standard Borel. An unknown index $\theta \in \Theta$ specifies a prepared law and positive normalized causal kernels (positivity means nonnegative measures, not a strictly positive density at every report). It is fixed for an entire execution; it may index an arbitrary history-dependent physical evolution. Policies see only past reports, legal physical commands and their own allowed coins. They never see θ. All assertions below are for the declared experiment and test, not for a formal parameter manifold.

A call has a nonnegative cost bounded below by $c_* > 0$. A predictable flag declares whether the call is productive. Commitment is allowed only after a productive report. Conditional on commitment, a fixed bounded future audit $Z \in H$ is executed, where $H \subset \mathbb{R}^k$ is compact and convex. This audit is not observed before the prediction is committed. The loss is $\|Z - r\|^2$, $r \in H$. Each productive call has a potential audit specified by the same kernel, whether or not the observer commits there. A fresh stopping coin is independent of that audit conditional on the current physical report and history. All other informative components of the current report must be included in the conditional laws below.

The cost N in the core theorem counts preparation, failed gate calls, productive calls, and all calls preceding commitment. A separately required final physical audit has its own explicit cost. Thus a protocol with one initial preparation call and one final audit uses total raw budget $N + 2$, not N. Time, bit operations and scratch space are not identified with raw-call count.

The stopping interface and retained labels. A checkpoint procedure may retain the entire precommit history, but its committed value belongs to one fixed dictionary of at most m elements. This convention counts only its terminal dictionary, not its full-history working memory. Its physical acquisition and audit costs are still charged.

A finite-state observer has an alphabet S of at most M labels and one fixed parameter-independent program. At every intercall boundary its only observer memory is the retained label. Its next physical command distribution is a fixed function of that label. In particular, the physical port identity, elapsed time, prior report and subprogram index are not additional free inputs. During a call the current command, its report and fresh independent coins are available; the post-report transition can depend on them. No such coin, report, real register or clock persists unless encoded in S. A fixed terminal decoder $r : S \to H$ reads only the committed label; it never reads the report or the audit. The two stopping interfaces are as follows.

Definition 2.1 (Event and state stopping)**.** In the *event-stopped* interface, a post-report kernel emits $(s', e) \in S \times \{0, 1\}$. The value $e = 1$ is permitted only after a productive call and commits exactly once: the experiment stops, the label s' is retained, and the decoder returns $r(s')$. No next command is executed. The commitment event belongs to the public stopped transcript; it is not an extra stored bit available during subsequent computation. On branches with $e = 0$, execution continues at s'. A label may therefore be used both live and terminal on different branches.

In the *state-stopped* interface there is instead a fixed map $h_* : S \to \{0,1\}$. The initial label has type zero. A transition commits if and only if $h_*(s') = 1$, and entry into type one is permitted only after a productive report. Type-one labels are globally terminal and absorbing: they have no executed command row. Type-zero labels are live. The halt status is thus completely determined by the retained state, and live and terminal labels are disjoint.

Let L_γ be the set of labels reached before physical calls and T_γ the set reached on commitment, under the relevant family of path laws. The program's role counts are $\ell_\gamma = |L_\gamma|$ and $t_\gamma = |T_\gamma|$, with overlap $o_\gamma = |L_\gamma \cap T_\gamma|$. Its actual alphabet count is $\ell_\gamma + t_\gamma - o_\gamma$, whereas its number of distinct decoder values is at most t_γ. A mandatory lower on ℓ_γ does not imply a subtraction from M unless overlap is excluded. This is the distinction used below, not an unmentioned memory convention.

Under robust almost-sure commitment and robust expected core cost at most N, write the infimal worst-case risks as $R^{\Theta}_{\mathrm{cp}}(N,m)$, $R^{\Theta}_{\mathrm{ev}}(N,M)$ and $R^{\Theta}_{\mathrm{st}}(N,M)$. An empty legal class has value $+\infty$. Ideal Borel programs may have fixed real constants; finite descriptions, evaluation time and scratch space are treated separately. A randomly selected program used for the entire execution is not a free persistent seed. Both stopping interfaces are legitimate resources, but neither is silently identified with the other.

3. Calibration measures on actual histories

The data in this section replace, rather than supplement, an exact admitted reference model. Fix a known positive prepared comparison experiment. Its productive report–audit law, in a declared mark coordinate, is a probability $K_0(dx, dz)$ on $X \times H$. Its mark marginal is ν. The comparison experiment is genuinely preparable, but need not belong to the operating family Θ. Its preparation, calibration observations and stored specification are separate resources; no uniform certificate below is inferred from a finite data sample without an additional statistical theorem.

Disintegration on standard Borel spaces gives Borel versions

$$f(x) = \int z K_0(dz \mid x), \quad w(x) = \int \|z - f(x)\|^2 K_0(dz \mid x), \quad d_r(x) = w(x) + \|f(x) - r\|^2. \tag{3.1}$$

They may be completed arbitrarily on a ν-null set, with $f \in H$. Put $L = \operatorname{diam}(H)^2$ and assume H is nonempty. For $b \geq 1$ and integer $m \geq 1$ define

$$\mathcal{Q}_m(b) = \min_{\substack{C \subset H,\ 1 \leq |C| \leq m \\ 0 \leq h \leq b,\ \int h\, d\nu = 1}} \int \big[w(x) + \operatorname{dist}(f(x), C)^2\big] h(x) \nu(dx). \tag{3.2}$$

Set this value to $+\infty$ when $b < 1$ or $m < 1$. Lemma 10.1 proves attainment. The case $L = 0$ has zero risk whenever a legal program exists and may be omitted from nonzero-rate assertions. The law ν is a comparison preparation law, not the law of reports accepted under an erroneous operating experiment. The transfer estimates quantify this difference.

A reference certificate without an exact reference model. Choose a single actual model $\theta_* \in \Theta$. For each productive pre-call history h and legal command a, let $P_{h,a}(dy, dz)$ be its conditional law of the *full visible current report* and the prospective audit. On the same report space let $P^0_{h,a}$ be a row of the declared comparison experiment, with a Borel mark $x = \pi_{h,a}(y)$. Its mark law is ν at every such row, and

$$\mathbb{E}^0[Z \mid y] = f(\pi_{h,a} y), \qquad \mathbb{E}^0[\|Z\|^2 \mid y] = w(\pi_{h,a} y) + \|f(\pi_{h,a} y)\|^2. \tag{3.3}$$

The conditional moments are given the full report, not merely the mark. All specified rows and defects are measurable in history and command. Versions are required on every legal history reachable under the actual witness, not only almost everywhere under an unrelated comparison path law. An independent uniform coin may be adjoined to y on both sides; the certificates below are stable under this adjunction by integration. Thus they cover randomized transitions and stopping.

Suppose nonnegative predictable bounds $a_-(h,a), e_-(h,a)$ satisfy, for every Borel $q : y \mapsto [0,1]$ and every finite-image Borel $r : y \mapsto H$,

$$\left| \int q(y)(P_{h,a} - P^0_{h,a})(dy, H) \right| \le a_-(h,a), \tag{3.4}$$

$$\left| \int q(y)\|z - r(y)\|^2 (P_{h,a} - P^0_{h,a})(dy, dz) \right| \le e_-(h,a). \tag{3.5}$$

These compare unnormalized task measures. In particular a joint probability total-variation bound a_- implies $e_- = La_-$. Bounds on order-zero, first and squared-norm measures on the *full report space*, as in Proposition 11.3, give a sometimes smaller e_-. Comparing the mark marginal alone is insufficient when the observer sees other informative components.

For every competing legal procedure under this *same actual witness*, require

$$\mathbb{E}_* C_\tau \ge \kappa_- \mathbb{E}_* A_\tau, \qquad \kappa_- > 0. \tag{3.6}$$

The number A_τ counts all productive calls through commitment. For each declared budget N let $\mathfrak{a}_N, \mathfrak{e}_N$ be certified bounds, uniformly over every competing procedure with robust cost at most N, on

$$\mathbb{E}_* \sum_{i \le \tau:\ i \text{ productive}} a_-(H_{i-1}, a_i), \qquad \mathbb{E}_* \sum_{i \le \tau:\ i \text{ productive}} e_-(H_{i-1}, a_i), \tag{3.7}$$

respectively. These are bounds on actual pre-call occupation, not on the occupation of a preferred algorithm. Uniform row bounds a_-, e_- give $\mathfrak{a}_N = (N/\kappa_-)a_-$ and $\mathfrak{e}_N = (N/\kappa_-)e_-$. Proposition 5.1 gives a less restrictive raw drift certificate. Mandatory live-state bounds used below are likewise bounds under the actual witness, not under the comparison experiment.

An approximately balanced common construction. Independently, suppose one parameter-independent acquisition program has B live roles and conditional expected cost at most κ_+ per attempt, on every surviving history in every actual model. It obtains one productive report, and returns to its start routing label after rejection. Within the attempt it forms a virtual mark x and a fresh choice J with exports $T_J : H \to H$. For every finite $C \subset H$ assume

$$|\{T_j r : j \in J, r \in C\}| \le D_0 |C|, \qquad D_0 \in \mathbb{N}. \tag{3.8}$$

All persistent routing for this operation is counted in B. Assume constants $0 \le a_+ < 1$, $e_+ \ge 0$, uniformly over every surviving history and actual model, such that

$$\| \operatorname{Law}(x \mid h) - \nu \|_{\mathrm{TV}} \le a_+, \tag{3.9}$$

$$\left| \mathbb{E}[q(x)\|Z - T_J r(x)\|^2 \mid h] - \int q(x) d_{r(x)}(x) \nu(dx) \right| \le e_+ \tag{3.10}$$

for all Borel $q \in [0,1]$ and finite-image Borel r. Total variation of probabilities is $\sup_D |\mu(D) - \nu(D)|$. The averaging includes the within-attempt coins and hidden history; no such coin may be retained for free. These are a construction certificate, not a restriction on competing strategies. Neither independent attempts nor a normalized-filter contraction is assumed.

4. The two-sided causal transfer theorem

For an interface $\iota \in \{\mathrm{cp}, \mathrm{ev}, \mathrm{st}\}$ write $R_\iota(N, M)$ for the risks of Section 2. Suppose every terminating state-stopped competitor needs at least $B_0 \ge 1$ live labels at θ_*. Let

$$m_-(M) = \begin{cases} M, & \iota = \mathrm{cp}, \mathrm{ev}, \\ M - B_0, & \iota = \mathrm{st}, \end{cases} \qquad m_+(M) = \begin{cases} \lfloor M/D_0 \rfloor, & \iota = \mathrm{cp}, \mathrm{ev}, \\ \lfloor (M-B)/D_0 \rfloor, & \iota = \mathrm{st}. \end{cases}$$

For the event upper additionally require $M \ge B$. If no stronger live lower is known, $B_0 = 1$ is always valid for a procedure initially waiting to make a physical call.

Theorem 4.1 (Calibration-stable causal acquired-geometry transfer)**.** *Assume precisely the experiment, row, effort, occupation and construction certificates of Sections 2–3; an exactly matching reference model in* Θ *is* not *assumed. Fix* N *and* M *with* $b_- = N/\kappa_- \geq 1$ *and* $m_-(M) \geq 1$*. Put* $t_N = (1 - \mathfrak{a}_N)_+$*. If* $t_N > 0$*, then*

$$R_\iota(N, M) \geq \Big[t_N \mathcal{Q}_{m_-(M)}(b_-/t_N) - \mathfrak{e}_N \Big]_+ . \tag{4.1}$$

When $t_N = 0$*, the asserted lower is zero, with no evaluation of* $\mathcal{Q}$ *at infinity. An empty legal class has risk* $+\infty$ *and is not made feasible by this formula.*

For $m_+(M) \geq 1$*, the stated event feasibility condition, and* $b_+ = N/\kappa_+ \geq (1 - a_+)^{-1}$*, one common Borel program at the* same *budget* N *attains an upper bound*

$$R_\iota(N, M) \leq (1 + b_+ a_+) \mathcal{Q}_{m_+(M)} \Big(\frac{b_+}{1 + b_+ a_+} \Big) + b_+ e_+ . \tag{4.2}$$

It commits almost surely, with at most B *live and* $D_0 m_+(M)$ *output roles. For* $1 \leq b_+ < (1 - a_+)^{-1}$ *replace the right side by* $\mathcal{Q}_{m_+(M)}(1) + e_+$ *using first-attempt commitment.*

In particular, with uniform lower defects a_-, e_- *and* $b_- a_- < 1$*, the lower reads*

$$\Big[(1 - b_- a_-) \mathcal{Q}_{m_-(M)} \Big(\frac{b_-}{1 - b_- a_-} \Big) - b_- e_- \Big]_+ . \tag{4.3}$$

At zero defects these inequalities recover the exact role-dependent identities of Theorem 9.1 whenever effort, live counts and identity exports match. The lower ranges over all legal strategies; the upper is one program for the entire actual model family.

The proof uses comparison rows along *actual* histories. A product coupling of whole stopped paths would require a separate termination and cost comparison and is not used.

Lemma 4.2 (Accepted comparison submeasure)**.** *Fix a legal competing procedure with committed outputs in a fixed dictionary* $C \subset H$*,* $|C| \leq m$*, and finite actual expected cost. Under the lower certificates, there are finite measures* ξ_c *on* X*,* $c \in C$*, such that, for* $\xi = \sum_c \xi_c$*,*

$$\xi \leq (\mathbb{E}_* A_\tau) \nu, \tag{4.4}$$

$$|\xi(X) - 1| \leq \mathfrak{a}_N , \tag{4.5}$$

$$\mathbb{E}_* \| Z - \widehat{r} \|^2 \geq \sum_{c \in C} \int d_c(x) \xi_c(dx) - \mathfrak{e}_N . \tag{4.6}$$

The measures need not be terminal laws of any comparison process.

Proof. Represent each current random transition by an independent uniform variable on $[0, 1]$, using the standard Borel kernel representation. Absorb it into the current report on both sides. Fix a reached productive history and the chosen command. The transition supplies an acceptance probability $q_h(y) \in [0, 1]$ and, on its acceptance event, a finite-image readout $r_h(y) \in C$. Equivalently one may use the acceptance indicator with the adjoined uniform variable. Complete r_h arbitrarily on rejection. Conditioning also on current command coins causes no loss: (3.4)–(3.5) hold for every chosen legal command.

For Borel $D \subset X$, define

$$\xi_c(D) = \mathbb{E}_* \sum_{i \leq \tau:\ i \text{ productive}} \int \mathbf{1}_{\{\pi_{H_{i-1}, a_i}(y) \in D\}} q_{H_{i-1}}(y) \mathbf{1}_{\{r_{H_{i-1}}(y) = c\}} P^0_{H_{i-1}, a_i}(dy, H).$$

The expectation is over actual pre-call histories and commands; it does not replace their distribution. All integrands are jointly measurable, so Tonelli defines finite measures. Summing over c and dropping $q_h \leq 1$ gives (4.4), since every comparison row has mark law ν. Actual commitment occurs exactly once almost surely. Its corresponding sum of actual conditional acceptance probabilities

therefore has expectation one. Subtract the two sums, apply (3.4) termwise, and use (3.7), obtaining (4.5).

On the actual disjoint commitment events the expected terminal losses sum to the risk. Replace only the current report–audit row in each contribution. The sum of absolute errors is at most $\mathfrak{e}_N$. By (3.3), the replaced conditional contribution is precisely $d_{r_h(y)}(\pi y)$. This proves (4.6). One may first truncate the number of calls; nonnegativity and the finite accumulated defect bounds justify passage to the limit. The comparison process need neither stop under this program nor satisfy its budget. □

Proof of Theorem 4.1. For the lower fix any admissible competitor. Its actual number of distinct outputs is at most $m_-(M)$: checkpoint and event-stopped procedures have at most M, whereas intrinsic state stopping leaves at most $M - B_0$ because its live and terminal alphabets are disjoint. This is a statement about its actual state labels, not about a fictitious comparison execution.

Use Lemma 4.2 with its fixed dictionary. If $t_N > 0$, put $t = \xi(X) \geq t_N$ and $\xi' = (t_N/t)\xi$. Then ξ' has mass t_N, and $\xi' \leq b_-\nu$ by the actual effort certificate. Thus $h = (d\xi'/d\nu)/t_N$ is a probability density with cap b_-/t_N. Nonnegativity of squared loss and nearest-dictionary projection give

$$\sum_c \int d_c \, d\xi_c \geq \int [w + \operatorname{dist}(f, C)^2] d\xi \geq t_N \mathcal{Q}_{m_-(M)}(b_-/t_N).$$

Combine this with (4.6), use nonnegativity, take the worst case over actual models, and then the infimum over all competitors. No minimax interchange or comparison-model membership is involved.

For the upper take an attained dictionary and density at cap $\beta = b_+/(1 + b_+a_+) \geq 1$, and accept the virtual mark with probability h/β. Its conditional actual acceptance probability on every reached attempt is at least $p = 1/\beta - a_+ > 0$. Its conditional unnormalized loss is at most $\mathcal{Q}_{m_+(M)}(\beta)/\beta + e_+$. Iterating the conditional acceptance bound gives probability at most $(1 - p)^j$ of reaching attempt $j + 1$. Summing conditional costs gives $\kappa_+/p = N$; summing the disjoint terminal loss contributions gives (4.2). This argument needs only (3.9)–(3.10), not a reference-model membership assumption. The packing construction in Lemma 10.5 uses $B + D_0 m$ labels for state stopping and $\max\{B, D_0 m\}$ for event stopping. A checkpoint counts only the latter output width. The first-attempt rule has acceptance one and needs no denominator. Uniform defects in (3.7) give (4.3) directly. This proves every assertion. □

Corollary 4.3 (Matching curves under two-sided calibration)**.** *Suppose Proposition 13.1 holds for the actually prepared comparison law, with fixed constants, and put* $F(b, m) = \max\{b^{-2/\alpha}, (bm)^{-2/(d+\alpha)}\}$. *Suppose* b_-/b_+ *and the positive widths* m_-/m_+ *remain bounded above and below. On a set of feasible budgets assume*

$$\mathfrak{a}_N \leq a_0 < 1, \qquad b_+a_+ \leq A_0 < \infty, \qquad \mathfrak{e}_N \leq c_0 F(b_-, m_-), \quad b_+e_+ \leq C_0 F(b_-, m_-).$$

For sufficiently small $c_0 > 0$*, depending only on the stated geometry,* a_0 *and the bounded effort and width ratios, one has*

$$R_\iota(N, M) \asymp F(b_-, m_-).$$

The comparison need not be an actual operating model. At a fixed nonzero calibration error the conditions need not persist to arbitrarily fine resolution; no exact dimension identity between perturbed and comparison supports is asserted.

Proof. Let $t_0 = 1 - a_0$. The lower geometry bound gives

$$t_N \mathcal{Q}_{m_-}(b_-/t_N) \geq c \min\{t_0^{1+2/\alpha}, t_0^{1+2/(d+\alpha)}\} F(b_-, m_-).$$

Take c_0 at most half this coefficient. The upper cap and width differ from b_-, m_- only by bounded positive factors. Apply the upper geometry bound to (4.2) and absorb the two error bounds. This proves constants uniform over the specified budgets. □

5. Nonuniform calibration and metastable occupation

A small uniform defect at every possible history is unnecessary. What the converse uses is the predictable defect accumulated before actual commitment, including rejected productive calls. The following criterion is stated at raw call boundaries and covers adaptive policies.

Proposition 5.1 (A drift certificate for accumulated defects)**.** *Let $d_i \ge 0$ be a predictable defect charge at call i, zero when that call is not productive. Suppose there are nonnegative integrable potentials V_i on a filtration containing the physical and observer history, with V_0 integrable, and a constant $\lambda \ge 0$ such that for every permitted current command, on $\{i \le \tau\}$,*

$$d_i + \mathbb{E}_*[V_i \mathbf{1}_{\{i<\tau\}} \mid \mathcal{F}_{i-1}, a_i] \le V_{i-1} + \lambda \mathbb{E}_*[c_i \mid \mathcal{F}_{i-1}, a_i]. \tag{5.1}$$

After termination set the potential to zero. Then every legal procedure with $\mathbb{E}_* C_\tau \le N$ has

$$\mathbb{E}_* \sum_{i \le \tau} d_i \le \mathbb{E}_* V_0 + \lambda N. \tag{5.2}$$

The potentials are analytic physical-state certificates. They are not additional memory available to the observer. The same assertion holds if the conditional expectation of a nonnegative physical defect dominates the visible-history defect; apply the bound to the physical defect and use the tower property.

Proof. For a deterministic k, multiply (5.1) by $\mathbf{1}_{\{i \le \tau\}}$, integrate, and sum from 1 to k. Conditional measurability of the decision to make the call makes all interior potential terms cancel. The remaining terminal term is $\mathbb{E}_*[V_k \mathbf{1}_{\{k<\tau\}}] \ge 0$. Drop it, bound the partial expected cost by N, and let $k \to \infty$ by monotone convergence. If commands are randomized, integrate the commandwise inequality first. The conditional-dominance version follows by applying the tower property to each nonnegative summand before summing. No optional stopping theorem with unverified uniform integrability is needed. □

Theorem 5.2 (Metastable calibration with explicit rare-mode mass)**.** *Suppose the actual reference contains a hidden mode $D_i \in \{0,1\}$ at raw call boundaries. For every actual history and legal command, its one-call transition obeys*

$$\mathbb{P}_*(D_i = 1 \mid D_{i-1} = 0, \mathcal{F}_{i-1}, a_i) \le u, \qquad \mathbb{P}_*(D_i = 0 \mid D_{i-1} = 1, \mathcal{F}_{i-1}, a_i) \ge v > 0.$$

Here $0 \le u \le 1$, $0 < v \le 1$, and $\pi_0 = \mathbb{P}_*(D_0 = 1)$ is the actual preparation mass. Every raw call costs at least $c_* > 0$. Conditional productive mass and task defects are bounded by the visible conditional expectations of $a_g + a_b \mathbf{1}_{\{D_{i-1}=1\}}$ and $e_g + e_b \mathbf{1}_{\{D_{i-1}=1\}}$, respectively. Let $b_- = N/\kappa_-$. Then Theorem 4.1 applies with

$$\mathfrak{a}_N = a_g b_- + \frac{a_b}{v}\pi_0 + \frac{a_b u}{v c_*} N, \tag{5.3}$$

$$\mathfrak{e}_N = e_g b_- + \frac{e_b}{v}\pi_0 + \frac{e_b u}{v c_*} N. \tag{5.4}$$

All competing adaptive stopping rules are covered. The bad-mode defects need not be small, and v may be small; their residence time and preparation mass remain explicitly charged. The matching geometric conclusion of Corollary 4.3 follows whenever these bounds satisfy its inequalities.

Proof. For the bad-mode mass charge use $V(D) = (a_b/v)\mathbf{1}_{\{D=1\}}$. In mode zero its conditional increase is at most $a_b u/v$. In mode one its conditional decrease is at least a_b, canceling the bad-mode charge. Hence

$$a_b \mathbf{1}_{\{D_{i-1}=1\}} + \mathbb{E}_*[V(D_i) - V(D_{i-1}) \mid \mathcal{F}_{i-1}, a_i] \le a_b u/v.$$

Killing the potential on commitment can only decrease the left side. Since $c_i \ge c_*$, Proposition 5.1 bounds the sum over all calls by $a_b \pi_0/v + a_b u N/(v c_*)$. Charging only productive calls is smaller. The good baseline contributes at most $a_g \mathbb{E}_* A_\tau \le a_g b_-$. This proves (5.3); replace a by e for (5.4).

Observations may reveal or bias the mode on surviving histories; conditioning and the drift estimate already include that selection. Neither stationary preparation nor independence of the stopping time from the mode is assumed. □

This is a substantive nonuniform class. For example take $a_g = e_g = 0$, a fixed bad-mode defect, $u = \eta v$, and initial mass $\pi_0 = \eta$, with $\eta, v > 0$. The rowwise worst defect remains fixed as $\eta \to 0$, while the actual accumulated bound is of order $\eta/v + \eta N/c_*$. Long bad episodes are permitted. The transient term cannot be silently removed by replacing actual preparation with stationary mass. Such a mode can modulate any positive report–audit instrument, including an atomic or changing-rank instrument; its mixture weights in this proof are the actual mode probabilities.

Proposition 5.3 (Selection amplifies reference error in both directions)**.** *For every $b \geq 1$ and $0 \leq e \leq 1/b$ there are positive prepared comparison and actual productive kernels with identical report laws, one unit of cost per call, and uniform task defect e, for which*

$$\mathcal{Q}_m(b) = be, \qquad R_{\mathrm{cp}}(b, m) = R_{\mathrm{ev}}(b, m) = 0$$

for all $m \geq 1$. The comparison need not belong to the actual singleton family. Reversing the two kernels gives $\mathcal{Q}_m(b) = 0$ and actual minimax risk be. Thus a term of order be is unavoidable on both sides of a general transfer statement, even with unrestricted terminal width. The appropriate state-stopped constructions use one additional live label.

Proof. Let each call draw an independent uniform report $x \in [0, 1]$, and let D have mass $1/b$. Audits are unobserved, conditionally mean zero and independent of prior history. Outside D, in both experiments the audit is a fresh fair sign. On D put a fresh fair sign of magnitude $\sqrt{be}$ in the comparison and put zero in the actual experiment. Both first moments vanish. For every $q \in [0, 1]$ and every report-dependent finite readout, the difference of unnormalized squared losses is $be \int_D q \, d\nu \leq e$. The comparison's residual variance is everywhere at least be, and selecting D with output zero attains that variance at expected cost b. Its trimmed profile is therefore be for every positive width. In the actual experiment the same rule has risk zero. One event-stopped state can reject by remaining at itself and commit by emitting a final event while retaining that same state; two intrinsically halt-typed states are required and sufficient. On reversing the kernels, every actual conditional residual variance is at least be, so every almost-surely committing strategy has risk at least be, and the same rule attains it. The audit amplitude, not its report marginal, has changed. This is an all-program argument, not a failure of one proposed sampler. □

6. Two raw-kernel verifications with imperfect references

We verify the same two-sided theorem in a classical controlled experiment and a noncommuting instrument. These examples extend the exact constructions of Section 14; the physical audit is the post-command task in each. The nonintervening sensing result of Theorem 16.1 has a different task and is retained separately.

Use the two-port gate of Lemma 14.1. All gate success probabilities are at least $p_0 > 0$. In the actual reference they are at most $p_1 \leq 1$. The command supports of the two ports are disjoint, with no free port input. Thus $B_0 = B = 2$, and the same predictable count argument gives

$$\kappa_- = 1 + 1/p_1, \qquad \kappa_+ = 1 + 1/p_0. \tag{6.1}$$

Indeed each productive call needs a successful gate call. Under the reference, the expected number of gate successes is at most p_1 times the expected number of gate calls, so total cost is at least $(1 + 1/p_1)\mathbb{E}_* A_\tau$. The gate-first construction has conditional expected cost at most $1 + 1/p_0$. No exact gate parameter is needed. Gate reports can reveal physical information; all subsequent conditional certificates below must still hold after that information.

Theorem 6.1 (A classical imperfect-reference realization)**.** *Let $E \in \{-1, 1\}$ be an arbitrary hidden sign at a productive call. Within that call the commands are identity and sign reversal, followed by a known nominal report–audit channel $J_E(dy, dz)$, $z \in H$. The declared comparison prepares a fair sign and has $K_0 = (J_1 + J_{-1})/2$; its full report is its mark, with derived moments* (3.1)*.*

Suppose the actual family contains a witness whose pre-command conditional sign bias, after every surviving history and legal productive command choice, has absolute value at most η_r. Each actual report–audit channel, conditional on either post-command sign and the full pre-call history, is within probability total variation ξ_- of J_E under this witness and within ξ_+ under every operating model. The common upper uses a fresh sign-reversal coin whose conditional law is within η_u of a fair coin, independent of the current hidden sign given pre-call history. Assume the gate conditions above.

Then the lower row certificates hold with

$$a_- = \eta_r + \xi_-, \qquad e_- = L(\eta_r + \xi_-),$$

and the common upper certificates hold with

$$a_+ = \eta_u + \xi_+, \qquad e_+ = L(\eta_u + \xi_+),$$

whenever $a_+ < 1$. Exports are the identity, so $D_0 = 1$. The two-sided bounds of Theorem 4.1 apply with actual witness effort (6.1)*. An exactly fair witness need not be present. In particular any nominal marked geometry verified from $(J_1 + J_{-1})/2$ yields matching curves in the certified regime of Corollary 4.3.*

Proof. For any probability weights on two signs, the variation distance between its mixture of J_1, J_{-1} and their fair mixture is at most the distance of the sign weights from the fair weights. Identity or reversal preserves the absolute sign bias. Applying the actual channel instead of J_E changes the full joint law by at most ξ_-, after integration over the sign. Therefore every chosen lower command and actual history has full report–audit distance at most $\eta_r + \xi_-$. This includes any auxiliary report through the declared channel; a report revealing a hidden audit coin would have to be included and satisfy this same bound. The bounded-loss argument gives e_-.

For the upper, write the hidden sign mean as $s \in [-1, 1]$ and the sign of the independent applied command as F, with $|\mathbb{E}F| \le 2\eta_u$. Then $|\mathbb{E}(EF)| \le 2\eta_u$, so the post-command sign law is within η_u of fairness, on every surviving history and for every hidden input distribution. Averaging the within-call command and using the channel error gives full task-law distance at most $\eta_u + \xi_+$. The algorithm accepts using the virtual report, not the discarded command coin. The audit is the post-command audit and its finite output is unchanged by export. Hence no command coin or hidden sign persists in the observer. The common controller has the two gate roles and the required output roles. All constants are independent of the chosen dictionary and budget. The gate argument already supplies effort and mandatory live roles; the general theorem now applies without using the realization to prove it. □

The reference condition can be verified, for example, by a genuinely biased preparation with parameter in $(0, \eta_r]$ at each productive call. The admitted witness is then not the fair comparison. More generally conditional biases and channel defects can depend on a hidden good/bad calibration mode. Convexity of variation gives the visible conditional defect bounds of Theorem 5.2. This realizes its nonuniform extension with actual correlated physical modes, not a change of the formal parameter dimension.

Theorem 6.2 (A noncommuting imperfect-reference realization)**.** *Let the productive physical system be a qubit. Its commands include the four Pauli conjugations, followed by a nominal normalized positive quantum instrument and its fixed prospective audit, with classical joint law $\mathcal{J}(\rho)(dy, dz)$, $z \in H$. The map $\mathcal{J}$ is affine, completely positive before its classical output, and trace preserving when all reports and audit outcomes are included. Take comparison state $\omega = I/2$ and $K_0 = \mathcal{J}(\omega)$.*

Suppose an actual witness satisfies $\frac{1}{2}\|\rho-\omega\|_1 \le \eta_r$ *conditional on every surviving physical history before a chosen productive Pauli command. The conditional actual full instrument, as a map to joint report and audit probabilities, differs from* $\mathcal{J}$ *by at most* ξ_- *in probability variation at this witness and at most* ξ_+ *uniformly on all states and histories in the operating family. Suppose the common upper's randomized Pauli channel* $\mathcal{T}$ *satisfies*

$$\sup_{\rho} \tfrac{1}{2}\|\mathcal{T}(\rho)-\omega\|_1 \le \eta_u.$$

Together with the same gate certificate, these raw conditions imply all the conclusions and constants of Theorem 6.1. No common eigenbasis, lower bound on posterior rank, or contraction of a normalized filter is required.

Proof. For a Hermitian trace-zero operator A, its positive and negative parts have equal trace $\|A\|_1/2$. A normalized positive instrument sends them to positive classical measures of those masses. Therefore $\|\mathcal{J}(\rho)-\mathcal{J}(\sigma)\|_{\mathrm{TV}} \le \|\rho-\sigma\|_1/2$. Pauli conjugation fixes ω and preserves trace norm. The witness bound and the primitive instrument error hence give lower full-joint distance at most $\eta_r+\xi_-$. For the upper, integrate the within-call Pauli choice first. Its average state is $\mathcal{T}(\rho)$, so contraction and the operating instrument error give distance at most $\eta_u+\xi_+$. Apply bounded squared loss to get the task defects. These comparisons concern unnormalized joint report–audit measures; they do not divide by a small outcome probability or claim continuity of a normalized state across a rank change.

Uniform Pauli weights give $\mathcal{T}(\rho)=I/2$. A weight vector at probability total-variation distance at most η_u from uniform gives the displayed bound, because it is a convex combination of density matrices. The audit follows the physical command, so exports are the identity. Its controller stores no qubit: the qubit belongs to the physical experiment, while the observer is the finite classical machine with the two explicitly charged gate roles. The remaining checks are the common gate count and actual full-report measurability. This proves the theorem directly from the raw instrument. □

For a concrete noncommuting family take the equatorial two-outcome instruments of Theorem 14.3, with continuously varying axis and effects $[I \pm \eta n\cdot\sigma]/2$, including both $\eta=1$ and $0<\eta<1$. The already derived comparison marked law, including its actual continuous and atomic weights, is unchanged as a calibration object. Biased near-mixed preparations and imperfect randomizations supply the new witness and construction errors. A hidden two-state mode can select a different positive instrument in bad periods; convexity gives the mode charges in (5.3)–(5.4). This allows genuine rank-changing and metastable rows. The geometry of the actual operating support is not asserted to equal that of the comparison: it is connected through the proved occupation and task bounds.

7. One-task composition with calibration and simulation

Define $\underline{Q}_m(N)$ and $\overline{Q}_m(N)$ to be the lower and upper expressions in Theorem 4.1 at a declared virtual or terminal width m, with the same effort and defect certificates; at $t_N=0$ put $\underline{Q}_m(N)=0$. These expressions are bounds, not new independently chosen geometric objectives.

Corollary 7.1 (An attained-error task with an imperfect reference)**.** *Take the calibrated class with the two-role gate, identity exports and a declared actual witness. Add one initial unit-cost observation of an independent fair sign* E_0*, erased with probability* 2ε*,* $0\le\varepsilon\le 1/2$*. The core kernels receive no information about* E_0 *other than its reported tag. They receive no information about a fixed unknown sign* $s\in\{-1,1\}$ *that influences only the last coordinate of the final unit-cost audit* $(Z,E_0,s\delta)$*,* $\delta\ge 0$*. All coordinates of the prediction are committed before that audit. Core certificates hold uniformly after every initial reported tag. For* $N'=N-2$ *in the feasible core range and*

$$m_{\mathrm{st}} = \lfloor (M-1)/3 \rfloor - 2 \quad (M\ge 13), \qquad m_{\mathrm{ev}} = \lfloor M/3 \rfloor \quad (M\ge 7),$$

one has, for either indicated interface,

$$\underline{Q}_M(N') + 2\varepsilon + \delta^2 \le R^{\Theta}_{\mathrm{joint},\iota}(N, M) \le \overline{Q}_{m_\iota}(N') + 2\varepsilon + \delta^2. \tag{7.1}$$

The lower uses terminal width M, which is safe for both interfaces, not a free extra dictionary. Under the matched-geometry and two-sided defect regime of Corollary 4.3, this yields

$$R^{\Theta}_{\mathrm{joint},\iota}(N, M) \asymp F(N'/\kappa_-, M) + \delta^2 + \varepsilon.$$

The early interface's causal deficiency to the perfect-sign interface is exactly ε, when its virtual sign must be delivered before any core call. The numerical and reference defects are separate from this attained erasure deficiency and from the unobservable calibration sign.

Proof. The tagged construction has one initial live role and three copies of the two core live roles. Its terminal roles are three copies of the core dictionary. Thus state stopping uses $7 + 3m$ labels, while event stopping uses $\max\{7, 3m\}$. Run the common calibrated upper in each tag, output the known sign or zero on erasure, and output zero for the calibration coordinate. Uniform conditional core certificates give the upper in (7.1) with exactly the core cost N'. No audit is available at prediction time.

For the lower fix the actual core witness and average the indistinguishable alternatives $s = \pm 1$. Conditional on every erased pre-audit history, including the stopping event, E_0 remains fair by causal induction: no core kernel reads the unrevealed sign. The erased coordinate therefore contributes at least 2ε. The averaged calibration-coordinate loss is at least δ^2. Under the same mixture, the core accepted comparison submeasure of Lemma 4.2 obeys the uniform row and occupation certificates and cost N', with at most M projected outputs. Its core risk is at least $\underline{Q}_M(N')$. These three bounds hold in the same risk calculation; different worst-case models are not added. The displayed width ratios are bounded on the stated ranges, proving the geometric consequence.

On erasure, any immediately produced virtual sign is independent of the hidden fair E_0. Thus the event that virtual and audited signs agree separates the perfect interface from every simulator by at least ε. Completing erased signs with an independent fair coin gives a disagreement coupling of probability ε, followed by the same causal continuation. This attains the reverse bound. This early-channel argument is the exact mechanism already proved in Theorem 18.3; its use here does not assume that the core has an exact admitted reference. The additional kernel independence assumptions prevent leakage of the erased sign through the core. □

Proposition 7.2 (Numerical execution at the prescribed physical budget)**.** *Fix a finite virtual dictionary, an intended acceptance function q, and the export readout $T_J r$. Suppose a finite description supplies a measurable implementation $\widetilde{q} \in [0, 1]$ and exported readout $\widetilde{r}_J \in H$ such that, on every physical report and within-call choice,*

$$|\widetilde{q} - q| \le \zeta_q, \qquad \|\widetilde{r}_J - T_J r\| \le \zeta_r.$$

The implementation uses the same number of declared output codes and no new intercall state. Its expected cost and risk bounds are given by the upper construction of Theorem 4.1 with

$$a_+ \text{ replaced by } a_+ + \zeta_q, \qquad e_+ \text{ replaced by } e_+ + L\zeta_q + 2\,\mathrm{diam}(H)\zeta_r.$$

Use the corrected design cap before evaluating q, so the final expected cost remains at most N. The certificate must be available uniformly for the selected corrected design. Program description, integration or threshold synthesis, temporary workspace, random-bit generation and evaluation time remain additional stated resources. Their finite bounds do not follow from Borel measurability.

Proof. The acceptance probability changes by at most ζ_q. For $z, r, r' \in H$, $|\|z - r\|^2 - \|z - r'\|^2| \le 2\,\mathrm{diam}(H)\|r - r'\|$. Thus each unnormalized task contribution changes by at most $L\zeta_q + 2\,\mathrm{diam}(H)\zeta_r$. The conditional-hazard summation in the central proof applies with these enlarged errors. The code indices and command routing are unchanged by hypothesis, so no extra labels have been granted. A description certifying the evaluation errors may be expensive; only Proposition 17.2's explicit

computational hypotheses, or a concrete implementation with the same guarantees, supplies the additional bit and time bounds. □

Complete stopped-law simulation composes with these estimates through Propositions 18.1–18.2: a bounded common loss incurs at most $L_{\mathrm{task}}\varepsilon_{\mathrm{sim}}$ in that direction, and actual expected cost needs its own certificate. Live simulator states and terminal export classes are charged before event-role reuse is allowed. Lower transfer through a different physical presentation requires a reverse resource-certified simulator or a separate actual witness; a one-way approximation is not a lower bound. The row defects above compare current productive tasks along actual histories and should not be renamed full-path deficiency.

8. Exact reference and balance certificates

The hypotheses in this section are additional exact conditions, used only in Theorem 9.1 and its stated applications. They are not imposed on Theorem 4.1.

A genuinely acquired reference. Fix one admissible $\theta_0 \in \Theta$ and a known, explicitly certified probability kernel $K_0(dx, dz)$ on $X \times H$. Under θ_0, after *every* legal productive command and every surviving pre-call history, the conditional mark law is $\nu(dx) = K_0(dx, H)$. Conditional on the full current report, the audit has the same first and second moments as $K_0(dz \mid x)$. Known within-call command coins or auxiliary reports supply no further audit information under this witness. Use the derived moments and profile of (3.1)–(3.2) for this actual reference law.

Assume that every legal observer satisfies, under this *same* witness,

$$\mathbb{E}_{\theta_0} C_\tau \geq \kappa_- \mathbb{E}_{\theta_0} A_\tau, \qquad A_\tau = \text{number of productive calls through commitment}, \tag{8.1}$$

where $\kappa_- > 0$ and C_τ is its actual accumulated core cost. This is an experiment-level all-strategy inequality. It may follow from a predictable compensated count or a bounded potential, and will be proved directly in the realizations. Since $A_\tau \geq 1$, $N < \kappa_-$ is infeasible under the witness.

One common upper program. Fix a parameter-independent acquisition program $\mathcal{G}$ using B live states. From its start state it obtains one productive report, almost surely, at conditional expected cost at most κ_+ on *every* surviving history and every θ. After rejecting that report it returns to the same program start. This is a reset of the observer's routing label, not an assumed physical reset of hidden dynamics.

Within the productive call, $\mathcal{G}$ is allowed a fresh finite choice $j \in J$, a Borel report map to a virtual mark x, and a Borel export map $T_j : H \to H$. No information about θ is used. For every pre-attempt history h, every θ, every Borel $D \subset X$ and every $r \in H$, require

$$\mathbb{P}_\theta(x \in D \mid h, \mathcal{G}) = \nu(D), \tag{8.2}$$

$$\mathbb{E}_\theta[\mathbf{1}_{\{x \in D\}} \|Z - T_j r\|^2 \mid h, \mathcal{G}] = \int_D d_r(x)\nu(dx). \tag{8.3}$$

The physical reference mark used in the lower and the virtual mark used in the upper may be different report functions; they take values in the same declared space and their compared law is the same actually prepared ν.

The averaging in these identities includes the fresh choice j, gate history inside the attempt, and all hidden states, but the rule later applied to the virtual mark may not retain the averaging coin except through the exported terminal value. The identities are verified from the raw kernels *before* selecting m, N or a dictionary. They neither require equality of the full hidden law nor assert equal conditional variances after conditioning on j.

There is a declared export multiplicity $D_0 \in \mathbb{N}$ such that

$$|\{T_j r : r \in C,\ j \in J\}| \leq D_0 |C| \quad \text{for every finite } C \subset H. \tag{8.4}$$

The important exact case is $T_j r = r$, $D_0 = 1$. Finite isometry groups give $D_0 \le |J|$, but are not required by the theorem. All routing needed to perform these operations between calls is already included in B. A random pulse performed in a separate call needs its own retained control state and cost.

The loss identity is an identity of unnormalized measures, not of posterior variances after averaging a normalized filter. It is equivalent to the corresponding order-zero, first and squared-norm moment identities when T_j is an isometry, as made precise in Proposition 12.1. It allows arbitrary report dependence and zero residual variance on individual actual histories.

The lower and upper effort constants need not agree. Exact risk identities below require the additional match $\kappa_- = \kappa_+ = \kappa$. In the raw heralding examples this match is proved by a predictable gate count. In applications without a match, the two distinct rescaled budgets remain in the conclusion.

9. The role-aware acquired-geometry theorem

Theorem 9.1 (Reference–balance transfer with explicit stopping roles)**.** *Assume the raw reference/effort and common task-balancing program conditions of Section 8. The reference and effort certificate use the same fixed admitted model. Write $b_- = N/\kappa_-$ and $b_+ = N/\kappa_+$.*

For every feasible lower budget,

$$\mathcal{Q}_m(b_-) \le R^{\Theta}_{\mathrm{cp}}(N, m), \tag{9.1}$$

$$\mathcal{Q}_M(b_-) \le R^{\Theta}_{\mathrm{ev}}(N, M), \qquad \mathcal{Q}_{M-1}(b_-) \le R^{\Theta}_{\mathrm{st}}(N, M). \tag{9.2}$$

For every $b_+ \ge 1$ and $m \ge 1$, one common Borel program gives

$$R^{\Theta}_{\mathrm{cp}}(N, D_0 m) \le \mathcal{Q}_m(b_+), \tag{9.3}$$

$$R^{\Theta}_{\mathrm{st}}(N, B + D_0 m) \le \mathcal{Q}_m(b_+), \qquad R^{\Theta}_{\mathrm{ev}}(N, \max\{B, D_0 m\}) \le \mathcal{Q}_m(b_+). \tag{9.4}$$

The latter program has at most B live roles and $D_0 m$ output roles. If every legal terminating observer at the same reference model needs at least B_0 live labels, then the state-stopped lower improves to $\mathcal{Q}_{M-B_0}(b_-)$. The event-stopped lower remains $\mathcal{Q}_M(b_-)$.

In particular, assume $\kappa_- = \kappa_+ = \kappa$, $B_0 = B$, and identity exports *$D_0 = 1$. Then the following attained equalities hold on the indicated feasible budgets:*

$$R^{\Theta}_{\mathrm{cp}}(N, m) = \mathcal{Q}_m(N/\kappa) \qquad (m \ge 1), \tag{9.5}$$

$$R^{\Theta}_{\mathrm{st}}(N, M) = \mathcal{Q}_{M-B}(N/\kappa) \qquad (M \ge B + 1), \tag{9.6}$$

$$R^{\Theta}_{\mathrm{ev}}(N, M) = \mathcal{Q}_M(N/\kappa) \qquad (M \ge B). \tag{9.7}$$

Here $N \ge \kappa$. Below κ every class is empty; the state-stopped class is empty for $M \le B$, and the event-stopped class for $M < B$. With nonidentity exports the width shifts in (9.3)–(9.4) *remain necessary in the statement. An optimal dictionary invariant under all exports gives its own unexpanded upper, but such invariance is an additional hypothesis, not automatic.*

The lower uses all competing strategies at one fixed model; the upper uses one program for all models and histories. The program need not simulate the full predictive state. Unequal effort constants leave a genuine interval $\mathcal{Q}(b_-) \le R \le \mathcal{Q}(b_+)$ at the appropriate widths; they do not give an exact budget identity. The role law does not assert an intrinsic minimal controller for an arbitrary physical interface. It gives exact minimax widths when the live lower and the constructed live upper match.

Corollary 9.2 (Regular orders under both stopping interfaces)**.** *Suppose the actual marked geometry obeys Proposition 13.1, effort is matched and the mandatory live count is B. For fixed export multiplicity D_0, define*

$$F(b, m) = \max\{b^{-2/\alpha}, (bm)^{-2/(d+\alpha)}\}.$$

For state-stopped observers with $M - B \geq D_0$, risk is comparable to $F(N/\kappa, M - B)$. For event-stopped observers with $M \geq \max\{B, D_0\}$, it is comparable to $F(N/\kappa, M)$. For checkpoint observers with $m \geq D_0$, it is comparable to $F(N/\kappa, m)$. Constants are independent of N, M, m and depend only on the stated geometry and export multiplicity.

Proof. The state-stopped lower width is $M - B$ and the virtual upper width is $\lfloor (M - B)/D_0 \rfloor$. The latter is at least $(M - B)/(2D_0)$. The event-stopped widths are M and $\lfloor M/D_0 \rfloor$, since their live roles can overlap the output roles. Apply Proposition 13.1; the checkpoint case is the same. Fixed unequal positive effort constants give analogous order bounds but not the exact equalities of the theorem. □

Atoms and zero-risk strata must be retained separately; Proposition 13.2 gives their exact mass threshold. None of these conclusions substitutes the dimension of a formal parameter set for an acquired covering or small-ball estimate.

10. Proof of the transfer and physical memory bounds

Lemma 10.1 (Attainment and continuity of the acquired profile)**.** *The minimum in* (3.2) *is attained. For each fixed m, $\mathcal{Q}_m$ is nonincreasing and left-continuous on $(1, \infty)$.*

Proof. Represent a dictionary by an ordered m-tuple in the compact space H^m, permitting repetitions. The feasible densities are a weak-star compact subset of the radius-b ball of $L^\infty(\nu)$: positivity, the upper bound and the mass equality are weak-star closed. Nets suffice; no unmentioned sequential compactness is used. The bounded loss functions vary uniformly continuously with the tuple, since squared distance on H is uniformly Lipschitz in its second argument. Along a convergent product net, split the objective difference into this uniform difference and the weak-star pairing with one fixed bounded measurable loss. This proves attainment. Monotonicity is immediate. If h is optimal at $b > 1$, use $h_\lambda = (1 - \lambda)h + \lambda$, whose cap is $b_\lambda = (1 - \lambda)b + \lambda < b$. Its loss tends to the optimal loss as $\lambda \downarrow 0$. Monotonicity sandwiches the values and proves the last assertion. □

Lemma 10.2 (A terminal measure obtained by any competitor)**.** *Under θ_0, every legal procedure committing exactly once, almost surely, of expected cost at most N has a terminal mark measure ζ satisfying*

$$\zeta(X) = 1, \qquad 0 \leq \zeta \leq (\mathbb{E}_{\theta_0} A_\tau)\nu \leq (N/\kappa_-)\nu. \tag{10.1}$$

If its terminal dictionary is C, its expected loss is at least $\int d_C \, d\zeta$, where $d_C = w + \operatorname{dist}(f, C)^2$.

Proof. For a Borel set D, predictability of productive calls and the conditional reference row imply, by Tonelli,

$$\mathbb{E}_{\theta_0} \sum_{t \leq \tau} \mathbf{1}_{\{t \text{ productive}\}} \mathbf{1}_{\{X_t \in D\}} = \mathbb{E}_{\theta_0} A_\tau \, \nu(D).$$

The terminal report is one of these reports, so its measure is dominated by this acquired report-count measure. Almost-sure commitment gives mass one, and (8.1) gives the last inequality. Only now take its Radon–Nikodym density, choosing a Borel version clipped to the displayed cap.

Condition on the entire current report, the preceding history and the chosen output. Under the witness, the conditional squared loss is $w(X) + \|f(X) - r\|^2$. The fresh stopping coin leaves this identity unchanged. Randomizing among a fixed dictionary cannot reduce the value below its nearest point. Summing over the disjoint commitment events proves the claim. No independence between different calls was assumed. □

Lemma 10.3 (One balanced rejection program)**.** *Let (C, h) be feasible in* (3.2)*. After each attempt of $\mathcal{G}$, accept virtual mark x with probability $h(x)/b$ and export $T_j r_C(x)$, where r_C is the first nearest point to $f(x)$ in a fixed ordering of C. Otherwise return to the start label. This program has expected core cost at most $\kappa_+ b$, risk exactly $\int d_C h \, d\nu$ under every θ, and at most B live roles and $D_0|C|$ output roles. It uses $B + D_0|C|$ state-stopped labels, or $\max\{B, D_0|C|\}$ event-stopped labels.*

Proof. Borel representatives and the finite nearest-point rule make the program measurable. On every surviving pre-attempt history, (8.2) gives acceptance probability $1/b$. For each nearest-point cell, apply (8.3) with its constant center, first to simple acceptance functions and then by bounded approximation to h/b. The expected terminal loss contributed by that attempt is

$$b^{-1} \int d_C(x) h(x) \nu(dx).$$

Consequently the probability of surviving j complete attempts is $(1 - 1/b)^j$. This is a consequence of the conditional hazard, not an assumed independent renewal structure. Summing the predictable attempt costs gives $\mathbb{E}C \le \kappa_+ \sum_{j \ge 0} (1 - 1/b)^j = \kappa_+ b$. Summing the terminal contributions gives the stated risk. For $b = 1$ the first attempt is accepted almost surely. The physical state need not return to its former distribution on rejection. Only the B routing labels and the exported terminal dictionary persist. The current choice j is discarded after its effect has been encoded in that terminal value or the routing label. Thus the role count follows from (8.4); Lemma 10.5 below gives the two alphabet counts. The conditional cost bound is for an entire attempt begun at its start history, not merely conditional on some favorable gate outcome. □

Proposition 10.4 (Semantic, presentation-invariant live-state bound)**.** *Consider finitely many mandatory pre-call contexts I with no additional free context input to the observer. A physical command includes its legal histories, executed kernel, port, visible reports and costs, not merely its printed name. Assume that the legal physical command set at every history in context i is exactly $\mathcal{U}_i$. Suppose that, under one fixed admitted model, every legal terminating procedure visits each context with positive probability. Let b_{phys} be the least number of subsets covering I such that each subset J has $\bigcap_{i \in J} \mathcal{U}_i \neq \varnothing$. Every autonomous observer needs at least b_{phys} live states. This count is invariant under duplicating command names or bijective stateless recoding of the same physical commands.*

Proof. For a live label that occurs in a collection J of mandatory contexts, its fixed command distribution must be supported on commands legal in every member of J. Null exceptional commands can be removed simultaneously because J is finite. Thus the intersection is nonempty. The collections served by its live labels cover the mandatory contexts, proving the bound. Duplicating a name changes neither the physical intersections nor this covering number; a stateless bijection preserves them. A translator that detects an unavailable phase is not stateless and must be charged as a simulator. The proposition does not characterize the full minimal automaton in general. □

Lemma 10.5 (Exact packing of live and output roles)**.** *A finite program with $\ell \ge 1$ live roles and $t \ge 1$ terminal roles can be implemented with $\ell + t$ state-stopped labels or $\max\{\ell, t\}$ event-stopped labels, preserving the complete stopped scored transcript and every path cost. Conversely, any event-stopped program with at most M labels has at most M terminal values. A state-stopped program with ℓ reached live labels has at most $M - \ell$ terminal values. These statements hold for randomized Borel transitions.*

Proof. For the state-stopped implementation use disjoint copies of the live and terminal role sets and type them zero and one. For the event-stopped implementation choose injections of the live roles and of the terminal roles separately into an alphabet of size $\max\{\ell, t\}$. On a continuing transition encode the next live role with the first injection. On a committing transition emit the final event and encode the output role with the second injection. The next command row is used only on continuing branches, and the decoder only after commitment. Thus their tables may share a label without conflict. No mode bit is stored: the event ends execution before any further command. Coupling the original and recoded within-call coins proves equality of every finite prefix and of the stopped transcript; almost-sure commitment completes the latter assertion. A label always has one fixed terminal decoder value, proving the converse dictionary bound. In the state-stopped model, the fixed halt map makes the reached live labels globally unavailable for terminal values, proving its subtraction. □

More generally, for any individual event-stopped program the exact bookkeeping identity is $|L_\gamma \cup T_\gamma| = \ell_\gamma + t_\gamma - o_\gamma$. If a declared interface permits at most r overlaps and requires at least B live roles, then

$$t_\gamma \le M - (B - r)_+.$$

Indeed $\ell - \min\{r, \ell\}$ is nondecreasing in $\ell \ge B$. When the constructed B-role program permits arbitrary such cross-role identifications, this bound is achieved at every feasible width by the injections in the proof. The state and event models are the cases $r = 0$ and unrestricted overlap, respectively. This is an overlap-aware resource statement about the entire transducer, not a consequence of counting live states alone.

Proof of Theorem 9.1. Lemma 10.2, applied at the fixed admitted θ_0, and minimization over the observer's fixed terminal dictionary prove the checkpoint lower. The event-stopped dictionary has at most M values by Lemma 10.5. A state-stopped observer has at least one live label, giving width $M - 1$, or at least B_0 when the same-witness physical certificate of Proposition 10.4 applies. This proves all converses without charging a live label twice or forbidding legal reuse.

Lemmas 10.1 and 10.3 give an attained optimizer and one common conditional-hazard implementation at cost $\kappa_+ b_+$. Apply the packing lemma to its B live and at most $D_0 m$ terminal roles. These give the independent upper assertions. Under matched effort, identity exports and $B_0 = B$, the lower and upper widths coincide in each of the three interfaces, yielding (9.5)–(9.7). With an invariant optimal dictionary the same construction has its actual orbit width; without it no unexpanded equality is asserted. The witness requires at least one productive call, so $N < \kappa_-$ is infeasible. The mandatory live count and the presence of a terminal value give the two different state thresholds exactly as in Definition 2.1. □

11. State-dependent approximate balance

Exact task balance can be replaced by a uniform conditional certificate. The reference and the all-strategy effort inequality are left unchanged. The upper program may now have arbitrary history-dependent physical states. Write $\nu_{\theta,h}$ for the virtual report law of an attempt begun at a surviving history h, and suppose

$$\|\nu_{\theta,h} - \nu\|_{\mathrm{TV}} \le a < 1, \tag{11.1}$$

$$\left|\mathbb{E}_\theta[q(X)\|Z - T_J r(X)\|^2 \mid h] - \int q(x) d_{r(x)}(x)\nu(dx)\right| \le e. \tag{11.2}$$

Here total variation between probabilities is $\sup_D |\mu(D) - \nu(D)|$. The second bound is required for every Borel $q : X \to [0,1]$ and every finite-image Borel $r : X \to H$, uniformly in θ, h. It compares the complete task measure, not just report marginals or unweighted conditional variances. The program has B live roles, export multiplicity D_0, and conditional expected attempt cost at most κ_+. The constants are certified before selecting a dictionary.

Theorem 11.1 (Approximate acquired-geometry transfer)**.** *Under the preceding conditions, for $\beta \ge 1$ with $\beta a < 1$, one common program uses at most B live and $D_0 m$ terminal roles and satisfies*

$$\sup_\theta \mathbb{E}_\theta C_\tau \le \frac{\kappa_+ \beta}{1 - \beta a}, \qquad \sup_\theta \mathbb{E}_\theta \|Z - \widehat{r}\|^2 \le \frac{\mathcal{Q}_m(\beta) + \beta e}{1 - \beta a}. \tag{11.3}$$

It commits almost surely. Thus the alphabet counts are $B + D_0 m$ for state stopping and $\max\{B, D_0 m\}$ for event stopping. For a specified $b = N/\kappa_+ \ge (1 - a)^{-1}$, put $\beta = b/(1 + ba)$. At the same physical budget N, the upper becomes

$$(1 + ba)\mathcal{Q}_m\left(\frac{b}{1 + ba}\right) + be. \tag{11.4}$$

For $1 \le b < (1-a)^{-1}$ the first-attempt rule instead has cost at most κ_+ and risk at most $\mathcal{Q}_m(1)+e$. All converses in Theorem 9.1 continue to hold at $b_- = N/\kappa_-$. No independence, renewal, or normalized-filter contraction is assumed.

Proof. Take an attained (C, h_*) at cap β, and accept with $q = h_*/\beta$. Its reference acceptance mass is $1/\beta$. By (11.1), its conditional physical acceptance probability on every surviving history is at least $p_- = 1/\beta - a > 0$. Consequently the probability of reaching attempt $j+1$ is at most $(1-p_-)^j$, by iterated conditioning. In particular commitment is almost sure and the expected number of attempts is at most p_-^{-1}. This proof does not condition on successful attempts.

The nearest-center rule is finite-image and measurable. By (11.2), the expected loss contributed by any reached attempt is at most $\mathcal{Q}_m(\beta)/\beta + e$. Sum the nonnegative terminal contributions over the disjoint commitment events, and sum conditional attempt costs over the surviving attempts. Both sums are bounded by their one-attempt upper times p_-^{-1}, proving (11.3). Lemma 10.5 supplies the alphabet counts. The substitution for β has $1 - \beta a = (1+ba)^{-1}$ and gives the displayed budget and risk exactly. The restriction on b is precisely $\beta \ge 1$. With $q = 1$, the first attempt always commits, regardless of the report perturbation; hence the small-budget assertion follows without dividing by an approximate acceptance mass. The converse proof uses only the admitted reference and was not altered. □

Corollary 11.2 (A stable geometric regime)**.** *Suppose the marked geometry of Proposition 13.1 holds, the effort constants have a fixed positive ratio, and the relevant role widths differ by at most a fixed export factor. If ba is bounded, b is above the feasibility threshold in Theorem 11.1, and*

$$be \le C_0 \max\{b^{-2/\alpha}, (bm)^{-2/(d+\alpha)}\},$$

then upper and lower risks are comparable to the same marked-geometry profile at b, m. These are sufficient stability conditions, not necessary error floors for every experiment.

Proof. The cap $b/(1+ba)$ differs from b by a bounded factor. Apply the two power terms of Proposition 13.1 to (11.4) and use the unchanged converse and fixed width factors. □

Proposition 11.3 (Primitive moment certificates)**.** *For identity exports, let μ_0, μ_1, μ_2 be the conditional report, vector first-moment and scalar squared-norm measures of an attempt. Suppose, uniformly over all models and surviving histories,*

$$|\mu_0 - \nu|(X) \le d_0, \quad |\mu_1 - f\nu|(X) \le d_1, \quad |\mu_2 - (w + \|f\|^2)\nu|(X) \le d_2,$$

where $|\cdot|$ denotes total variation of scalar or finite-dimensional vector measures. If $\sup_{r\in H} \|r\| \le R_0$, then (11.1)–(11.2) *hold with*

$$a \le d_0/2, \qquad e \le d_2 + 2R_0 d_1 + R_0^2 d_0.$$

In particular, a conditional total-variation bound a on the joint report–audit law gives $e = La$ whenever all squared losses are bounded by L.

Proof. The two report measures are probabilities, so their probability total variation is half the variation norm of their difference. Expand the loss as $\|Z\|^2 - 2\langle r(X), Z\rangle + \|r(X)\|^2$ and integrate the three bounded coefficient functions against the indicated signed measures. This proves the first assertion for every q, r simultaneously. The last assertion applies the defining probability total-variation bound to a measurable function taking values in $[0, L]$. Vector coordinates may be taken in the affine hull of H; no ambient redundant coordinates are needed. □

Proposition 11.4 (Sharp rare-selection scales)**.** *The factor b multiplying the one-attempt task defect in* (11.4) *cannot be replaced by a universal smaller-order factor, even with equal report laws and one live role. Likewise the waiting denominator $1 - \beta a$ in* (11.3) *is attained by a zero-loss selection problem. Absence of first-moment balance can prevent every common causal program, not merely a proposed finite mixture, from attaining the reference profile.*

Proof. For the first assertion take $H = [-1, 1]$, and let every productive call cost one and draw an independent mark uniformly on $[0, 1]$. Let D have mass $1/b$. In the reference model the audit is zero on D and is a fresh fair sign off D. In a second model it is a fresh fair sign multiplied by $\sqrt{be}$ on D and an ordinary fair sign off D, where $0 \le be \le 1$. Audit coins are unobserved before commitment. Both first moments vanish, all reports agree, and for every acceptance rule and response the difference of unnormalized losses is

$$be \int_D q(x)\nu(dx) \le e.$$

The reference profile is zero, attained by selecting D and predicting zero. In the second model every conditional residual variance is at least be, so every almost-surely committing program has risk at least be, regardless of its memory. Selecting D commits in expected b calls and attains be uniformly over the two models. Thus the common minimax risk is exactly be.

For the denominator take reference report mass $1/\beta$ on a marked region D, with zero audit there and a fresh fair sign outside. Perturb its mass to $1/\beta - a > 0$ by moving mass outside D. This changes the report law by exactly a. Every zero-risk rule must commit in D, and the reference-count argument at the perturbed independent law forces expected calls at least $(1/\beta - a)^{-1}$. First entry into D attains this value. The case $\beta = 1$ can be treated with two report atoms, the second initially null. This establishes sharp waiting amplification, not a universal positive risk penalty for mere report error.

Finally consider three models with exactly the same constant pre-audit reports and no task-changing commands. Their audits are respectively $0, +1, -1$. The reference profile at the first model is zero at every feasible budget. For any common program, the law of its committed output R is the same under the last two models. Their average loss is $1 + \mathbb{E}R^2 \ge 1$. The constant output zero attains worst risk one. This is a complete indistinguishability proof for all legal history-dependent programs. It is a limitation of that particular information structure, not a claim that general causal balance is impossible. The underlying decision-theoretic argument is classical. □

12. Raw balance and stability without a variance floor

A finite moment criterion. Let E be a finite-dimensional real ordered space with generating positive cone and a strictly positive mass functional u. Its normalized positive base $S = \{s \ge 0 : u(s) = 1\}$ is compact. This is a verification device for the following proposition; Theorem 9.1 does not assume a finite-dimensional physical space.

For each of finitely many allowed within-call commands j, suppose the raw joint report–audit law depends linearly and positively on $s \in S$. After the declared report transformation, and a task isometry $T_j : H \to H$ with $T_j H = H$, write $\widetilde{Z} = T_j^{-1} Z$. The following vector measures are linear functionals of s:

$$A_j(D)(s) = \mathbb{E}_s[\mathbf{1}_{\{x \in D\}}], \quad B_j(D)(s) = \mathbb{E}_s[\mathbf{1}_{\{x \in D\}} \widetilde{Z}], \quad C_j(D)(s) = \mathbb{E}_s[\mathbf{1}_{\{x \in D\}} \|\widetilde{Z}\|^2].$$

Choose an origin in $\mathrm{aff}(H)$ and orthonormal coordinates on its $k_0 = \dim \mathrm{aff}(H)$-dimensional direction space. Here B_j has k_0 coordinates; the written squared norms are in these translated coordinates. Translation leaves all compared squared losses unchanged. Fix a genuinely executed reference law K_0 and its moments (3.1).

Proposition 12.1 (Raw task-moment balance criterion)**.** *A fixed distribution $\lambda \in \Delta(J)$ balances report and squared-loss measures for every normalized input s if and only if, as E^*-valued measures,*

$$\sum_j \lambda_j A_j(dx) = \nu(dx)u, \tag{12.1}$$

$$\sum_j \lambda_j B_j(dx) = f(x)\nu(dx)u, \tag{12.2}$$

$$\sum_j \lambda_j C_j(dx) = (w(x) + \|f(x)\|^2)\nu(dx)u. \tag{12.3}$$

Equality on a countable determining algebra of report events suffices. The set of such λ is compact and convex. If it is empty, finitely many scalar moment constraints already separate all mixtures in $\Delta(J)$ from the required identities.

Proof. Expand $\|\widetilde{Z} - r\|^2 = \|\widetilde{Z}\|^2 - 2r \cdot \widetilde{Z} + \|r\|^2$. The three identities give (8.2)–(8.3), using that T_j is an isometry. Conversely, the mass identity and equality of this quadratic polynomial for r in the affine hull of H determine the first moments in that affine hull and the squared-norm moment. One may first translate the affine hull to the origin and use an orthonormal coordinate system on it; orthogonal coordinates of Z are then fixed and carry no additional constraint. Equality for all $s \in S$ is equality as linear functionals because the positive cone is generating. Uniqueness of finite signed measures extends equality from a determining algebra to all Borel sets. Each coordinate constraint is affine and closed in the finite simplex. Compactness and the finite-intersection property give a finite inconsistent subfamily if no mixture exists. Its affine image of the simplex is compact and convex and omits the prescribed moment vector. Finite-dimensional strict separation proves the last assertion. This is a certificate of impossibility for the declared command family, not for arbitrary causal algorithms. □

The criterion is insensitive to density zeros or changes in the rank of a posterior. It also shows why report-only averaging is insufficient. A report identically equal to zero has the same law for hidden states $e = \pm 1$, but an audit $Z = e$ has loss $(e - r)^2$. Without balancing its first moment there is no common d_r valid for all outputs, despite perfect report balance.

A finite physical randomization satisfying $\sum_j \lambda_j G_j(s) = \omega u(s)$ is one sufficient source of (12.1)–(12.3) for any subsequent fixed instrument. More generally, it suffices to annihilate input dependence only on the report, first-moment and squared-norm functionals; the entire hidden state need not be erased. These are linear moment identities, not an assertion that a Fisher matrix, covariance or fixed-rank parametrization is primitive. Randomization itself is standard [3], [4]; the stopped conclusion follows only after the actual witness, paid acquisition and exported readout conditions have also been verified.

Integrated predictive-update stability. The useful continuity is that of unnormalized instruments. Uniform pointwise continuity of a normalized posterior can fail at zero-mass reports.

Proposition 12.2 (Integrated stability through singular reports)**.** *Let a classical stochastic instrument, or a finite-dimensional completely positive trace-preserving instrument, map two normalized inputs s, t to subnormalized post-report states A_x, B_x with respect to a common finite dominating measure for these two inputs. This measure may be chosen pairwise; the inequalities are unchanged by replacing it with any equivalent dominating representation. Put $p_x = u(A_x)$, $q_x = u(B_x)$, and*

normalize on positive-mass reports. In the classical ℓ^1 norm or quantum trace norm,

$$\int \|A_x - B_x\|_1 \, dx \le \|s - t\|_1, \tag{12.4}$$

$$\|p - q\|_{\mathrm{TV}} \le \tfrac{1}{2}\|s - t\|_1, \tag{12.5}$$

$$\int \min(p_x, q_x) \left\| \frac{A_x}{p_x} - \frac{B_x}{q_x} \right\|_1 dx \le 2\|s - t\|_1. \tag{12.6}$$

Zero-mass terms are defined to contribute zero. The same inequalities apply to a complete finite causal continuation instrument under the same fixed parameter-independent policy. If its stopped scored instrument terminates almost surely under both inputs, the unnormalized contraction and report bound extend to that stopped instrument.

Proof. In the quantum case write the Hermitian difference $s - t = D_+ - D_-$ with positive parts. Positivity, the triangle inequality and preservation of integrated trace give $\int \|\mathcal{I}_x(D_+ - D_-)\|_1 dx \le \operatorname{tr} D_+ + \operatorname{tr} D_- = \|s - t\|_1$. The same Jordan-decomposition argument works for classical signed measures. Taking masses proves (12.5). When $p \le q$,

$$p\|A/p - B/q\|_1 = \|A - (p/q)B\|_1 \le \|A - B\|_1 + (q - p).$$

Interchange A, B for the other case and integrate, using $|p - q| \le \|A - B\|_1$. Composition of legal physical instruments remains positive and trace preserving, so the finite-continuation assertion follows. For stopped paths, sum over commitment time, first to a finite horizon while retaining the surviving output. This is a positive mass-preserving instrument. Almost-sure termination under both inputs lets the surviving masses tend to zero; monotone summation gives the stopped assertion. No bound on expected stopping time follows from this argument; it requires a separate cost certificate. □

This stability controls the *actual report mass* together with the posterior, including pure states and likelihood zeros. It is not a contraction factor strictly below one, and no normalized posterior is stored by the finite-state upper. In particular a pointwise variance-floor constant diverging along nearly pure histories does not force the risk constant of Theorem 9.1 to diverge.

13. Actual marked geometry, singular supports and atoms

The following calculation is inherited from the stopped-experiment and predictable-geometry results [1], [2]. It is included to make the sharp consequences independent of those manuscripts.

Let $\sigma = (f, \sqrt{w})_{\#}\nu$, supported on $K \times [0, 1]$ with compact $K \subset H$. Suppose there are $d, \alpha > 0$, $0 < r_0 \le 1$ and constants $c, C > 0$ such that for every $x \in K$, $0 < r \le r_0$ and $0 < s \le 1$,

$$c r^d s^\alpha \le \sigma(B(x, r) \times [0, s]) \le C r^d s^\alpha. \tag{13.1}$$

The estimates are global on the stated component K; a bound only near one point is not enough. These are statements about the *actual* report–task pushforward, not the volume of a reachable parameter set. They include Ahlfors-regular singular location measures of noninteger d and a reliability coordinate with exponent α.

Proposition 13.1 (Marked acquired profile)**.** *Under* (13.1)*, for all $b \ge 1$ and $m \ge 1$,*

$$\mathcal{Q}_m(b) \asymp \max\{b^{-2/\alpha}, (bm)^{-2/(d+\alpha)}\}, \tag{13.2}$$

with constants depending only on the displayed geometric data and the bounded location set.

Proof. A finite cover of K by balls of radius $r_0/2$ gives $\sigma(K \times [0, s]) \le C_0 s^\alpha$. Under any density cap b, the selected reliability obeys $\mathbb{P}(\sqrt{w} \le s) \le b C_0 s^\alpha$. Choose $s = (2C_0 b)^{-1/\alpha}$, decreasing it by a fixed factor if necessary, to obtain the first lower bound. For any dictionary of m points, the sublevel $w + \operatorname{dist}(f, C)^2 \le t^2$ is contained in m rectangles of location radius t and reliability at most t. Centers outside K can be moved for this covering argument to a point of K within t, increasing

the radius to $2t$; empty balls are discarded. For $2t \le r_0$ its mass is at most $C_1 m t^{d+\alpha}$. Taking $t = c_1(bm)^{-1/(d+\alpha)}$ with a sufficiently small fixed c_1 bounds the selected sublevel mass by $1/2$ and gives the second lower. This choice works for all $b, m \ge 1$ after reducing c_1 once.

For the upper, the marginal $\sigma(\cdot \times [0,1])$ satisfies two-sided d-dimensional ball bounds. Maximal separated sets therefore provide at least $c_2 r^{-d}$ centers at separation $4r$ for small r. Choose

$$j = \min\{m, \lfloor c_3 b^{d/\alpha} \rfloor\}, \qquad t = C_2(bj)^{-1/(d+\alpha)}.$$

Take $c_3 > 0$ sufficiently small and C_2 sufficiently large, in that order with the packing inequality also enforced by decreasing c_3. For $b \ge b_0$, where

$$b_0 = \max\{1, (2/c_3)^{\alpha/d}, (C_2/\min(r_0/8, 1))^{d+\alpha}\},$$

we have $j \ge 1$, $t \le \min(r_0/8, 1)$, and j disjoint rectangles $B(x_i, t) \times [0, t]$ whose union has mass at least $cjt^{d+\alpha} \ge 1/b$. Here the required packing condition $jt^d \le c_4$ follows from $j \le c_3 b^{d/\alpha}$. A density of mass one bounded by b can be supported on this union (constant inverse union mass suffices), and its distortion to these centers is at most $2t^2$. Substituting j gives the upper in (13.2). For $1 \le b < b_0$, accept every report and use any fixed point in H. This gives a uniform upper while $b^{-2/\alpha} \ge b_0^{-2/\alpha}$ supplies a uniform positive comparison, independently of unbounded m. No compactness assertion about the integer m is needed. □

Proposition 13.2 (An exact acquired atom transition)**.** *Suppose the marked law is $\sigma = \lambda\delta_{(x_*,0)} + (1-\lambda)\sigma_0$, $0 < \lambda < 1$, where σ_0 is a probability law and has no atom at $(x_*, 0)$. Write $\mathcal{Q}_m^0$ for its profile. If $b\lambda \ge 1$, $\mathcal{Q}_m(b) = 0$ for every $m \ge 1$. If $r = 1 - b\lambda > 0$, then for every $m \ge 2$,*

$$r\mathcal{Q}_m^0\left(\frac{b(1-\lambda)}{r}\right) \le \mathcal{Q}_m(b) \le r\mathcal{Q}_{m-1}^0\left(\frac{b(1-\lambda)}{r}\right). \tag{13.3}$$

Consequently the exact case of Theorem 9.1 retains the zero-risk threshold $N = \kappa/\lambda$, with no unknown mass renormalization.

Proof. Above threshold, select only the actual atom and output x_*. Below threshold, any selected probability has remainder mass $s \ge r$. With its fixed dictionary, scale its remainder submeasure by r/s. This yields a submeasure of mass r, still dominated by $b(1-\lambda)\sigma_0$, and with no larger nonnegative distortion than the original remainder part. Its minimum possible distortion over dictionaries of size m is the left side of (13.3). For the upper, select the full allowable atom mass $b\lambda$, allocate one center to x_*, and use an optimal $(m-1)$-point dictionary on a mass-r submeasure of the remainder. The unnormalized factor r is retained throughout. For $m = 1$ below threshold the lower $r\mathcal{Q}_1^0(b(1-\lambda)/r)$ is still valid. There is no zero-point upper dictionary. A valid one-point upper is obtained by fixing its center at x_* and minimizing $r\int[w + \|f - x_*\|^2]h\, d\sigma_0$ over $0 \le h \le b(1-\lambda)/r$, $\int h\, d\sigma_0 = 1$. It need not coincide with the unrestricted one-point optimum. □

The atom result applies to any remainder, including a mixture of singular strata. It does not assert that arbitrary changing-rank geometries satisfy (13.1). In the raw quantum construction below, rank changes can occur while the task law still satisfies these rectangle estimates; this is proved directly rather than inferred from ambient matrix dimension.

14. Two raw realizations with singular predictive states

Paid, history-dependent heralding. There are two physically distinct ports. At port G a gate call costs one. Its success probability, conditional on every hidden and observed past, belongs to $[p_0, 1]$, where $p_0 > 0$ is a known guaranteed lower bound, not the unknown true rate. Success permits one call at acquisition port A, costing one. No commitment is legal at G. After rejecting at A, the protocol returns to G. No hidden state is reset by this routing convention. The class includes a reference model with constant gate success p_0 and the productive laws specified below.

Lemma 14.1 (The exact robust paid-attempt scale)**.** *For this interface set $\kappa = 1 + 1/p_0$. Every all-history policy under the reference model satisfies $\mathbb{E}C = \kappa \mathbb{E}A_\tau$. The program which calls the legal current port reaches one productive report at conditional expected cost at most κ, for every model and history. Both ports require distinct live states, so $B = 2$ is a semantic mandatory lower. Robust expected-budget feasibility is exactly $N \geq \kappa$ with at least three state-stopped labels, or at least two event-stopped labels. Below those respective alphabet thresholds the legal classes are empty.*

Proof. Let G_τ, A_τ be gate and productive counts. Every productive call consumes one gate success, and a terminating path has no unused success. Predictable counting gives $\mathbb{E}A_\tau = p_0 \mathbb{E}G_\tau$ under the witness, without requiring independent cycles. Hence $\mathbb{E}C = (1 + 1/p_0)\mathbb{E}A_\tau$. For the upper, the probability of j consecutive failures, conditional on any history at the start, is at most $(1 - p_0)^j$. Summing gives at most $1/p_0$ gate calls and one acquisition call. The physical gate and acquisition ports have disjoint legal command sets, and both occur before any legal commitment under the same witness. Proposition 10.4 applies. Accepting the first productive report gives feasibility with one output role. Lemma 10.5 gives three state-stopped or two event-stopped labels, while the two mandatory live contexts give the converse thresholds. The decoder in the two-label construction may share a label with a live command; the emitted event prevents a further call. At $p_0 = 1$ the cost is two, not one. At fixed N, the task is infeasible when $1 + 1/p_0 > N$; rare-rate uniformity always refers to the feasible rescaled budget N/κ. □

Controlled classical hidden states. Let the physical state contain a sign $E \in \{-1, 1\}$ and an arbitrary nuisance state. Between productive calls any causal stochastic or deterministic evolution is allowed, with no mixing or positivity requirement. The observer may choose either of two physical commands on the sign: identity or sign flip, followed immediately by the same known observation-and-audit instrument. There is no unknown intervening evolution between this chosen command and that instrument. The physical task is to predict the *post-command* audit, not a pre-intervention sign.

For input sign e, let $J_e(dx, dz)$ be any fixed probability kernel with bounded audit in H; it can have atoms, singular parts and sign-dependent report likelihoods. Nuisance evolution is unrestricted provided this productive instrument depends on the sign through these stated kernels. The class contains a reference member which supplies a fresh fair sign before each productive command, independently of all prior reports, and uses the constant gate rate p_0. This is an admitted physical preparation model, not a reset available freely to every observer. Its productive law is

$$K_0 = \tfrac{1}{2}(J_+ + J_-). \tag{14.1}$$

Theorem 14.2 (Classical task balance without a transition or variance floor)**.** *The classical family satisfies Theorem 9.1 with $B = 2$, $D_0 = 1$ and $\kappa = 1 + 1/p_0$, for the task geometry derived from* (14.1)*. In particular, for $N \geq \kappa$, $M \geq 3$ in the first equality and $M \geq 2$ in the second,*

$$\begin{aligned} &R^{\Theta}_{\mathrm{st}}(N, M) = \mathcal{Q}_{M-2}(N/\kappa), \quad R^{\Theta}_{\mathrm{ev}}(N, M) = \mathcal{Q}_M(N/\kappa), \\ &R^{\Theta}_{\mathrm{cp}}(N, m) = \mathcal{Q}_m(N/\kappa) \qquad\qquad (m \geq 1). \end{aligned} \tag{14.2}$$

The constants require no positive lower bound on transition probabilities or conditional residual variances. The strictly positive gate floor p_0 remains an explicit resource assumption.

Proof. The upper chooses identity or flip with a fresh fair coin within the productive command. For every conditional pre-command sign distribution, averaging makes the post-command sign fair. By linearity the joint report–audit law, not merely its report marginal, is exactly K_0. Take the identity export map. The nuisance state need not be reset or stored. Under the reference member, each chosen physical command, even conditioned on its chosen coin and on the full preceding history, still has law K_0; hence arbitrary competitors satisfy the reference conditions. Lemma 14.1 gives both cost directions and the mandatory states. Theorem 9.1 proves (14.2). Classical instrument continuity follows from Proposition 12.2, including deterministic transitions. □

For a concrete singular geometry, draw (Y,T) independently of the sign with any probability ν_0 on $K\times[0,1]$, and let $Z=Y+TEe_1$ in a fixed compact convex carrier. The reference has $f=Y, w=T^2$. Choose a d-regular singular measure on K and reliability law $\mathbb{P}(T\le s)=s^\alpha$ to obtain (13.1); add a genuinely prepared atom $(Y,T)=(x_*,0)$ to obtain Proposition 13.2. The report can instead be informative: a weak sign report V with $\mathbb{P}(V=v\mid E=e)=(1+\eta ve)/2$ gives $f=Y+\eta VTe_1$ and $w=(1-\eta^2)T^2$ under the reference. Its profile is still exact, although rectangle estimates must then be checked for that sheared marked law rather than assumed. In contrast, a static known-to-the-history sign can have zero conditional variance at individual histories. Such histories violate a positive pointwise variance mark but do not invalidate this minimax theorem.

Noncommuting quantum instruments with rank changes. Consider a qubit with arbitrary unknown completely positive trace-preserving evolution between port interactions. No depolarizing component, spectral gap or full-rank floor is imposed. Gate instruments have success effects bounded below by $p_0 I$; their surviving states may be pure. The productive port permits a Pauli command from $\{I,\sigma_x,\sigma_y,\sigma_z\}$, followed by a fixed known instrument and audit, with no unknown operation inserted inside this command. The class contains a reference member that prepares $I/2$ immediately before every productive Pauli command and has gate success p_0.

Here is an explicit noncommuting instrument. Draw (y,t) with any fixed probability ν_0 on $K\times[0,1]$. Choose a Borel angle $\phi(y,t)$ and $0\le\eta(y,t)\le 1$. Set

$$n=(\cos\phi,\sin\phi,0),\qquad E_v=\tfrac12(I+v\eta n\cdot\sigma),\quad L_v=\sqrt{E_v},\qquad v=\pm1.$$

The unnormalized post-report state is $L_v\rho L_v$, and the report is $x=(y,t,v)$. At commitment measure σ_z, with result $e=\pm1$, and score $Z=y+tee_1$. The axes may vary, so effects at different reports need not commute.

Theorem 14.3 (Quantum verification through pure posterior strata)**.** *For the stated quantum family, the same conclusions* (14.2) *hold with the actual reference mark law* $\nu=\nu_0\otimes\frac12(\delta_{-1}+\delta_1)$ *and* $f=y, w=t^2$. *They remain valid at* $\eta=1$, *at pure incoming states, and across report strata where the posterior rank changes. If* ν_0 *has the geometry or atom described above, the corresponding regular or exact atomic risk law follows from the* same *central theorem.*

Proof. Writing $\rho=(I+r\cdot\sigma)/2$ shows directly that

$$\tfrac14\sum_{U\in\{I,\sigma_x,\sigma_y,\sigma_z\}}U\rho U^*=I/2.$$

This standard Pauli averaging identity [4] is a charged within-call preparation, not free additional quantum memory. In this call convention the pulse and subsequent instrument form one productive call; pulse execution time and precision are charged within it. An interface requiring a separate pulse call must add that call and its intervening state. Every unnormalized report–audit probability is linear in ρ, so its average is its value at $I/2$. The reference member is invariant under *each* chosen Pauli command, including after conditioning on that command and all earlier observations. Thus it gives the required reference for all competitors, while the random command gives one common upper for every unknown input state.

Under $I/2$, the probability of v is $\mathrm{tr}(E_v)/2=1/2$ and the normalized post-report state is E_v. Its z component is zero, so the subsequent Pauli audit has both signs with probability $1/2$. This gives the stated f,w and the full unnormalized loss identity. The eigenvalues of E_v are $(1\pm\eta)/2$. Choosing $\eta=1$ on one set of positive ν_0-mass and $\eta=1/2$ on another gives actual rank-one and rank-two strata, not merely a formal rank-deficient limit. Different equatorial axes give noncommuting effects. Proposition 12.2 supplies report continuity and actual-mass-weighted update stability without dividing by a positive lower eigenvalue. Lemma 14.1 and Theorem 9.1 complete the proof. □

The quantum carrier here is finite dimensional and the observer is classical. It has two live labels and its terminal dictionary, not an uncharged quantum register. Interventions change the declared post-command audit; neither theorem asserts that randomizing an unknown pre-intervention quantity preserves its original estimation task. Known classical or quantum preparation operations, their costs and the availability of the reference member are explicit hypotheses, not automatic consequences of positivity.

Corollary 14.4 (Imperfect controls through zero-mass and pure strata)**.** *In either realization, replace the exact uniform within-call randomization by a predictable distribution at probability total-variation distance at most a from uniform, independently of the current physical input conditional on the pre-command history. The distribution may depend on the model and preceding history through a physical implementation error. Keep the same gate certificate and the admitted reference preparation. Then the same raw class verifies Theorem 11.1 with $e = La$, where $L = \operatorname{diam}(H)^2$. The reference converse is unchanged. No transition, residual-variance, or eigenvalue floor is introduced.*

Proof. For the classical control, the post-command sign distribution is within a of fair, uniformly over its input distribution: subtract the two convex mixtures and use contraction of total variation. The subsequent fixed report–audit kernel cannot increase this distance. For the quantum control, writing $u_j = 1/4$ and actual weights λ_j gives

$$\left\| \sum_j \lambda_j U_j \rho U_j^* - I/2 \right\|_1 \le \sum_j |\lambda_j - u_j| \le 2a.$$

Any complete positive report–audit instrument contracts trace distance to classical probability total variation, as in Proposition 12.2. In both cases the joint conditional law is therefore within a of K_0. Apply Proposition 11.3. At the reference, every individual control preserves the fair sign or $I/2$; the fresh reference preparation and the stipulated independence of the mixing weights from that current input preserve the same all-strategy witness. This independence cannot be omitted for a fault that senses the unobserved input. The argument is entirely in unnormalized measures and remains valid on the singular acquired components already described. □

15. A nonintervening Gaussian sensing law

In this section the task is not randomized by a physical intervention. At a productive call a current hidden sign $E \in \{-1, 1\}$ emits

$$X = aE + \sigma G, \qquad a, \sigma > 0, \tag{15.1}$$

where G is a fresh standard normal variable. The audit is that current sign E. All history-dependent hidden sign evolutions are allowed, including the static signs and an independently refreshed fair sign. No command may alter the sign between the sensor call and its audit. The heralding interface is that of Lemma 14.1; all three indicated sign models with constant gate rate p_0 are admitted. Thus arbitrary old observations or gate histories may be informative, but the class has no positive uniform conditional variance floor.

Under the independent fair reference, the actual productive density and conditional moments are

$$\nu(dx) = \frac{\varphi_\sigma(x-a) + \varphi_\sigma(x+a)}{2} dx, \quad f(x) = \tanh(kx), \quad w(x) = \operatorname{sech}^2(kx), \quad k = a/\sigma^2. \tag{15.2}$$

The formulas follow by dividing the two Gaussian likelihoods; their ratio is $\exp(2kx)$. For $b \ge 1$, let $t_b \ge 0$ be the unique threshold with $\nu(|X| \ge t_b) = 1/b$, taking $t_1 = 0$, and define

$$D(b) = b \int_{|x| \ge t_b} w(x)\nu(dx), \qquad U(b) = 2b \int_{|x| \ge t_b} (1 - |f(x)|)\nu(dx). \tag{15.3}$$

The two-sided threshold construction and the logarithmic waiting scale were already proved for a positive-prior-floor class in [1]. Here the lower is instead supplied by one admitted fair reference

member: this removes that floor, permits static hidden signs, and yields a ratio-one minimax asymptotic uniformly over every larger terminal alphabet. The conditional-variance selection principle below is standard in regression with rejection [6], with the earlier recognition tradeoff in [5]. The assertion here additionally constrains the paid stopped experiment and all of its persistent states; it does not assert priority for a threshold rule.

Theorem 15.1 (Terminal-value transition and the sharp waiting asymptotic)**.** *Put $b = N/\kappa$, with $\kappa = 1 + 1/p_0$, and assume $N \geq \kappa$. One terminal value gives worst risk one. In particular,*

$$R^{\Theta}_{\mathrm{st}}(N, 3) = R^{\Theta}_{\mathrm{cp}}(N, 1) = 1. \tag{15.4}$$

For each R in the collection

$$\{R^{\Theta}_{\mathrm{st}}(N, M) : M \geq 4\} \ \cup \ \{R^{\Theta}_{\mathrm{ev}}(N, M) : M \geq 2\} \ \cup \ \{R^{\Theta}_{\mathrm{cp}}(N, m) : m \geq 2\},$$

one has $D(b) \leq R \leq U(b) \leq 2D(b)$, and uniformly over this entire collection,

$$R \sim 4\exp(-2a^2/\sigma^2)\exp\left[-\frac{2a}{\sigma}\sqrt{2\log(N/\kappa)}\right] \quad \text{as } N/\kappa \longrightarrow \infty. \tag{15.5}$$

The state-stopped class is empty below three labels; the event-stopped class is empty below two. Two event-stopped labels already permit both live ports and both terminal values. The asymptotic is not a fixed-budget limit as $p_0 \downarrow 0$.

Proof. The lower uses the independently refreshed fair sign, not a chosen posterior prior for each competitor. Lemma 10.2 bounds the terminal X-law by $b\nu$, and the conditional residual variance in (15.2) is unavoidable even with infinitely many output values. Since w decreases with $|x|$, its minimum integral against a mass-one density bounded by b is $D(b)$. For completeness, subtract the threshold rule from any feasible density: on $|x| \geq t_b$, $w(x) \leq w(t_b)$ and $h - b \leq 0$; on the complement, $w(x) \geq w(t_b)$ and $h \geq 0$. Integrating the product and using zero total mass difference proves this minimization without a restriction to threshold competitors.

For the upper, accept exactly when $|X| \geq t_b$ and output $\mathrm{sign}(X)$. Reflection of the Gaussian kernel shows that its acceptance probability is $1/b$ for each fixed current sign, and its unnormalized squared-loss contribution is $U(b)/b$ for each sign. The same identities hold for every mixture of signs conditional on any earlier history. The proof of Lemma 10.3 therefore gives expected cost at most κb and risk $U(b)$, without refreshing or estimating the hidden sign. This uses two live port roles and the two outputs ± 1, packed into four state-stopped or two event-stopped labels by Lemma 10.5. Algebra gives

$$\frac{2(1 - |f|)}{1 - f^2} = \frac{2}{1 + |f|},$$

so $D(b) \leq U(b) \leq 2D(b)$ and, when $t_b > 0$, $U(b)/D(b) \leq 2/(1 + \tanh(kt_b))$.

If there is only one terminal value $r \in [-1, 1]$, the two admitted static-sign models give worst-case risk at least $\max\{(1 - r)^2, (-1 - r)^2\} \geq 1$, irrespective of stopping. Outputting zero at the first productive report attains one. The mandatory physical ports leave only one terminal value for a state-stopped observer when $M = 3$. This subtraction is not valid for the event-stopped observer, whose two labels can carry the two different roles on different branches. This argument allows all fresh randomized stopping and routing, not only thresholds.

It remains to identify the asymptotic constant. Let $u_b = (t_b - a)/\sigma$ and $c = a/\sigma > 0$. The tail equation is

$$b^{-1} = \overline{\Phi}(u_b) + \overline{\Phi}(u_b + 2c).$$

Integration by parts gives, for $u > 0$, $\frac{u}{1+u^2}\varphi(u) \leq \overline{\Phi}(u) \leq \varphi(u)/u$. The second tail is negligible relative to the first, so $u_b = \sqrt{2\log b} + O(\log\log b/\sqrt{\log b})$. In particular $t_b = a + \sigma\sqrt{2\log b} + o(1)$. For any fixed $v > 0$, the same Gaussian tail bounds show $\nu(|X| > t_b + v)/\nu(|X| \geq t_b) \to 0$. Thus $|X| - t_b \to 0$ in probability under the selected tail law. Moreover $0 \leq w(X)/w(t_b) \leq 1$ there, and this

ratio tends to one in that probability. Bounded convergence in probability yields $D(b)/w(t_b) \to 1$. Since $w(t_b) \sim 4e^{-2kt_b}$, the right side of (15.5) follows. Finally $U(b)/D(b) \to 1$. The lower and upper do not depend on the available dictionary width once it is at least two, proving the claimed uniformity under the three listed budget conventions. □

This application is also a raw transformed-task verification of Theorem 9.1. After a sensor report, choose a fresh fair $g \in \{-1, 1\}$, use virtual pair (gX, gE) and export $T_g r = gr$. Averaging yields exactly the reference joint law for every pre-sensor sign distribution, with $D_0 = 2$. No physical randomization of the target occurs. The symmetric threshold rule above removes even the virtual coin. Its sharp asymptotic arises from the actually acquired Gaussian tails, not a formal one-dimensional parameter space. The one-value obstruction requires the two static models in addition to the fair lower witness; it is not inferred from a variance-floor hypothesis.

16. Nonintervening directional sensing

The preceding binary sensor has only two possible targets. We now keep a continuum of targets and obtain a joint waiting–memory law without a balancing control. Fix $d \geq 2$, $a, \sigma > 0$. Behind the two-port heralding gate of Lemma 14.1, a productive call observes

$$X = aU + \sigma G_d, \qquad U \in \mathbb{S}^{d-1}, \qquad G_d \sim N(0, I_d). \tag{16.1}$$

The current direction U is the terminal audit, unchanged between this observation and its audit; the Gaussian noise is fresh. The direction may change arbitrarily between calls, depending on hidden and observed history. The uncertainty class includes every static direction and the model with a fresh uniform direction at every productive call, with gate success p_0. There are no commands that change the current target. The prediction space is the closed unit ball. The decoder is fixed and finite, as before. The effort constant is $\kappa = 1 + 1/p_0$.

Let ν_d be the actual report law under the uniform-direction reference, $R = \|X\|$, and $V = X/\|X\|$ away from its null origin. Set

$$A_d(s) = \frac{\int_{-1}^{1} z e^{sz}(1 - z^2)^{(d-3)/2}\, dz}{\int_{-1}^{1} e^{sz}(1 - z^2)^{(d-3)/2}\, dz}, \quad s = aR/\sigma^2. \tag{16.2}$$

For $b \geq 1$, let $t_b \geq 0$ be the unique radial threshold with $\nu_d(R \geq t_b) = 1/b$, taking $t_1 = 0$, and define

$$D_d(b) = b \int_{R \geq t_b} [1 - A_d(aR/\sigma^2)^2]\, d\nu_d, \tag{16.3}$$

$$U_d(b) = 2b \int_{R \geq t_b} [1 - A_d(aR/\sigma^2)]\, d\nu_d. \tag{16.4}$$

The relevant output geometry is the covering radius of the actual audit support:

$$\rho_m = \min_{\substack{C \subset \overline{B}(0,1) \\ 1 \leq |C| \leq m}} \sup_{u \in \mathbb{S}^{d-1}} \operatorname{dist}(u, C). \tag{16.5}$$

This definition allows interior as well as surface centers; it is not an ambient-volume quantization surrogate.

Theorem 16.1 (Waiting and finite output geometry)**.** *For the raw experiment* (16.1)*, write* $b = N/\kappa \geq 1$*. For each of the three following risks, with its indicated available output width* m*,*

$$R = R^{\Theta}_{\mathrm{cp}}(N, m), \qquad R = R^{\Theta}_{\mathrm{ev}}(N, m)\ (m \geq 2), \qquad R = R^{\Theta}_{\mathrm{st}}(N, m + 2),$$

one has

$$\max\{D_d(b), \rho_m^2\} \leq R \leq \min\{1, (\sqrt{U_d(b)} + \rho_m)^2\}, \qquad D_d(b) \leq U_d(b) \leq 2D_d(b). \tag{16.6}$$

In particular, with constants depending only on d, a, σ,

$$R \asymp (1+\log b)^{-1/2} + m^{-2/(d-1)}. \tag{16.7}$$

For $m \le d$ the risk is exactly one at every feasible acquisition budget. For $m \ge d+1$ it is strictly below one at sufficiently large b, and for each fixed m it converges to ρ_m^2 as $b \to \infty$. Moreover,

$$D_d(b) \sim U_d(b) \sim \frac{(d-1)\sigma}{a\sqrt{2\log b}}. \tag{16.8}$$

The ratio of R to this last expression tends to one uniformly along any budgets with $\rho_m = o((\log b)^{-1/4})$; it suffices that $m/(\log b)^{(d-1)/4} \to \infty$. An event-stopped gate is infeasible below two labels, and a state-stopped gate below three. No unknown-sensor learning or intervention on U is asserted.

Proof. We give the raw-kernel, statistical and geometric steps separately.

Acquired posterior and the converse. Bayes' formula for the uniform sphere preparation gives conditional density proportional to $\exp((a/\sigma^2)\langle X, u\rangle)$ with respect to surface probability. Integrating its axial coordinate gives (16.2); transverse coordinates cancel by reflection. Thus the conditional mean is $A_d(s)V$ and the residual variance is $1 - A_d(s)^2$. These quantities were derived from the joint observation–audit kernel. Differentiation under its bounded axial integral gives $A_d'(s) = \mathrm{Var}_s(z) > 0$, while $A_d(0) = 0$. The residual variance is therefore decreasing in R.

The same-witness gate count and Lemma 10.2 imply that the terminal report measure of any competing rule has density at most b relative to ν_d. Forgetting its nonnegative dictionary loss gives the lower obtained by putting density b on the smallest residual variance, namely the radial tail of mass $1/b$. To see this directly, write $w_* = 1 - A_d(at_b/\sigma^2)^2$ and compare any feasible h with $b\mathbf{1}_{R \ge t_b}$: the integral of $(w - w_*)(h - b\mathbf{1}_{R \ge t_b})$ is nonnegative on both the tail and its complement, and the mass difference integrates to zero. This proves $R \ge D_d(b)$ against every legal strategy.

For a fixed dictionary C, a static direction u has loss at least $\mathrm{dist}(u, C)^2$ on every commitment. Taking the worst static u and then minimizing dictionaries gives the second lower in (16.6). It uses a different admitted model from the radial converse; hence the two lower contributions are combined by their maximum, not by an unjustified sum. Lemma 10.5 and the two mandatory gate contexts give exactly the stated dictionary widths.

A common nonintervening upper. At each productive call accept if $R \ge t_b$. Rotation invariance of the Gaussian noise makes the conditional acceptance probability $1/b$ for every current direction, and hence for every mixture determined by a surviving history. The expected cost is at most κb, by conditional hazards and the gate bound. The same rotation invariance makes

$$\mathbb{E}[\|U - V\|^2 \mid R \ge t_b, U = u]$$

independent of u. Average this identity over a uniform u and use the derived posterior mean to identify its value as $U_d(b)$. It remains valid after an arbitrary hidden history and after summing the disjoint commitment events. Choose a minimizing dictionary in (16.5), whose existence follows by compactness, and output its first nearest point to V. Pointwise $\|V - r_C(V)\| \le \rho_m$. The triangle inequality in L^2 gives risk at most $(\sqrt{U_d(b)} + \rho_m)^2$. The constant output zero gives the alternative upper one. The construction has two live roles and m output roles; apply the packing lemma. Finally $1 - A^2 = (1 - A)(1 + A)$ with $0 \le A \le 1$ proves $D_d \le U_d \le 2D_d$.

Noise asymptotics. Put $v = (d-3)/2 > -1$ and change variables $y = s(1 - z)$ in the denominator and first centered moment in (16.2). After removing their common factors, their integrands are $e^{-y}y^v(2 - y/s)^v$ and y times that function, on $[0, 2s]$. On $[0, s]$ dominated convergence applies, including $v = -1/2$. On $[s, 2s]$, the possible endpoint singularity is integrable and its contribution, including either of these two polynomial factors, is bounded by a polynomial in s times e^{-s}. The

limiting ratio is therefore $\Gamma(v+2)/\Gamma(v+1) = v+1$. Consequently

$$1 - A_d(s) \sim \frac{d-1}{2s}, \qquad 1 - A_d(s)^2 \sim \frac{d-1}{s}.$$

For fixed $U = u$, $a + \sigma G_1 \le \|au + \sigma G_d\| \le a + \sigma\|G_d\|$. One-dimensional Gaussian tail bounds and the polar integral for the χ_d tail sandwich the radial tail. They give

$$t_b = a + \sigma\sqrt{2\log b} + O\left(\frac{\log\log b}{\sqrt{\log b}}\right).$$

The same bounds show, for every $c > 0$, that the conditional probability of $R > (1+c)t_b$ given $R \ge t_b$ tends to zero. Thus $t_b/R \to 1$ in conditional probability and, since it is bounded by one, in conditional mean. The displayed posterior asymptotics are uniform over $R \ge t_b \to \infty$. Inserting them into (16.3)–(16.4) proves (16.8). On compact b intervals D_d is continuous and strictly positive, so $D_d(b) \asymp (1+\log b)^{-1/2}$ for all $b \ge 1$.

Finite output geometry. For $0 < r < 1/2$, the surface probability of a sphere intersected with any Euclidean ball of radius r, whose center may be inside the unit ball, is at most $C_d r^{d-1}$. Indeed choose one point in that intersection; the intersection lies in the sphere cap of chord radius $2r$ about that point. Polar surface coordinates bound that cap. Hence m such balls cannot cover the sphere unless $mr^{d-1} \ge c_d$. Conversely a maximal r-separated sphere set has at most $C'_d r^{-(d-1)}$ points, because its disjoint chord-radius $r/2$ caps each have mass at least $c'_d r^{d-1}$; maximality makes it an r-cover. Rescaling r and enlarging constants for bounded m gives $\rho_m \asymp m^{-1/(d-1)}$. Together with (16.6) and $(x+y)^2 \le 2x^2 + 2y^2$, this proves (16.7).

If $m \le d$, choose a unit vector orthogonal to the span of $c_i - c_1$ and orient it so that $\langle u, c_1\rangle \le 0$. Then every $\|u - c_i\|^2 \ge 1$. The center zero shows $\rho_m = 1$. For $m = d+1$, take regular simplex vertices u_i on the unit sphere, with $\langle u_i, u_j\rangle = -1/d$ for $i \ne j$. Their simplex contains the ball of radius $1/d$, since each facet lies at that distance from zero. Thus $\max_i \langle u, u_i\rangle \ge 1/d$ for every unit u. The centers $c_i = u_i/d$ consequently give $\rho_{d+1}^2 \le 1 - 1/d^2 < 1$. The assertions about exact risk one and eventual strict improvement follow from the two bounds. For fixed m, those bounds squeeze risk to ρ_m^2. Finally, when $\rho_m = o(\sqrt{D_d(b)})$, the upper divided by $D_d(b)$ tends to one, while the lower is at least one. This proves the uniform leading-constant assertion and completes the proof. □

The sphere posterior in this proof is the familiar exponential directional family; the derivation is included to fix the actual kernel and avoid an assumed information metric. The contribution here is the all-strategy waiting converse together with an unmodified-target implementation and an output-covering converse under the two fully specified stopping resources. It is distinct from choosing one report out of a fixed independent batch for distributional simulation, as in [15], [16].

17. Finite descriptions and stopped numerical certificates

An arbitrary Borel acceptance set is not a finite computer program. We first state the error bound on the scale that a stopped experiment actually uses.

Proposition 17.1 (Relative-acceptance implementation)**.** *Consider an implemented attempt program with the same legal physical ports and B live states as the ideal one. On every surviving pre-attempt history suppose its conditional expected cost is at most $\widetilde{\kappa}$, its acceptance probability is at least $(1-\rho)/b$, and its unnormalized terminal loss is at most $(Q+\eta)/b$, where $0 \le \rho < 1$, $\eta \ge 0$, $b \ge 1$. It then terminates almost surely and satisfies*

$$\mathbb{E}C \le \frac{\widetilde{\kappa} b}{1-\rho}, \qquad R \le \frac{Q+\eta}{1-\rho}. \tag{17.1}$$

The terminal alphabet is the actual exported dictionary. These bounds remain valid for unknown history-dependent implementation errors satisfying the stated conditional certificates.

Proof. The probability of surviving j attempts is at most $(1-(1-\rho)/b)^j$. Predictable summation of the bounded conditional costs and nonnegative terminal-loss contributions proves both inequalities and finite expected attempts. Each attempt terminates almost surely by its own cost certificate. No comparison of terminal distributions was used to infer duration. □

For example, an absolute one-attempt acceptance error e and unnormalized loss error e_L give $\rho = be$ and $\eta = be_L$, not $\rho = e$ and $\eta = e_L$. Rare selection amplifies these absolute errors. A task-readout perturbation of size δ in a carrier of diameter D_H changes loss by at most $2D_H\delta + \delta^2$ *conditional on acceptance*; its unnormalized error is weighted by acceptance mass. A continuous report and a digitized atomic report need not be close in total variation; boundary-event or unnormalized moment estimates must be supplied instead.

Proposition 17.2 (Finite-description upper at the same cost)**.** *In addition to exact balance, assume the known reference measure admits an effective sequence of finite physical report partitions with certified approximation of bounded acceptance functions in $L^1(\nu)$. Assume finite points in H and their exported values can be approximated within H with certified uniform error, and their primitive report classifiers and probabilities have declared finite descriptions, execution costs and integration error bounds. Then, for every $b \geq 1$, $m \geq 1$ and $e > 0$, there is a finite-description observer with at most $B + D_0 m$ states, cost at most κb and risk at most $\mathcal{Q}_m(b) + e$. This is an existence and approximation result under effective certificates, not a polynomial-time discovery theorem for arbitrary kernels.*

Proof. For $b > 1$, choose $1 < b' < b$. By Lemma 10.1, $\mathcal{Q}_m(b')$ can be made arbitrarily close to $\mathcal{Q}_m(b)$. Approximate its optimal finite dictionary inside H and use its nearest-center cells. Approximate the bounded acceptance h/b' and this finite center assignment jointly on a finite partition. The assumptions supply certified $L^1(\nu)$ error bounds. Because all squared losses are bounded and dictionary errors change them uniformly, the acceptance and unnormalized loss errors tend to zero. The ideal acceptance $1/b'$ has a strictly positive margin above $1/b$. Take the approximation sufficiently accurate that its acceptance is still at least $1/b$. Balance makes these quantities the same on every surviving history and model. Its risk is its unnormalized loss divided by acceptance; it tends to the ideal risk. The attempt count proof then gives cost at most κb. For $b = 1$, accept the first report identically and approximate only the dictionary and assignment; no acceptance margin is needed. Current partition indices and command coins are scratch variables, and only the routing or exported output label persists.

The argument is constructive once the indicated approximation and integration certificates are supplied. A finite search over candidate dictionaries and report partitions, with bounded integration intervals, can certify an available upper. No claim that a local optimizer supplies a global lower is involved. Fully effective global convergence additionally uses effective compactness of the dictionary carrier and uniform integration bounds; a finite dictionary net and the Lipschitz loss then give finite lower and upper profile approximations, potentially at exponential search cost. □

For regular geometry, finitely supplied centers and coordinate certificates give a direct upper without learning an arbitrary acceptance density. Replace inner rectangles from Proposition 13.1 by outward rounded rectangles, accepting every true inner rectangle and only points with marked distance at most a fixed multiple of its radius. Actual mass remains at least $1/b$ and distortion remains of the same order. With m centers in a fixed k-dimensional box and radius t, $p = O(1 + \log(1/t))$ precision bits suffice when the coordinate evaluation has that error certificate. A sequential implementation uses $O(mkp)$ read-only data bits, $O(kp+\log m)$ scratch bits and $O(mkp^2)$ elementary arithmetic operations per productive report, in addition to the specified physical evaluation and command costs. These are sufficient costs, not optimal bit/time lower bounds.

An end-to-end dyadic example without a variance floor. Use the classical family of Theorem 14.2 with independent uniform $(Y, T) \in [0,1]^2$, post-command audit $Z = Y + TE$, and $p_0 = 1/4$.

Thus $\kappa = 5$. Use three state-stopped labels: gate, acquisition, and terminal value $1/2$. The same one-output program packs into two event-stopped labels, but this does not prove that one output is optimal at the latter budget. Ideally accept

$$A = [3/8, 5/8] \times [0, 1/8].$$

Under balanced task measures,

$$\nu(A) = 1/32, \qquad \int_A [(y - 1/2)^2 + t^2]\, dy\, dt = 1/3072. \tag{17.2}$$

Let $e = 2^{-14}$, and suppose the physical digitizer supplies clipped coordinates with errors at most e, using only (Y, T) and not the discarded sign-flip coin. Accept when

$$3/8 + e \le \widehat{Y} \le 5/8 - e, \qquad \widehat{T} \le 1/8 - e.$$

The accepted true report lies in A, and every point in $A_{\mathrm{in}} = [3/8 + 2e, 5/8 - 2e] \times [0, 1/8 - 2e]$ is accepted. Put

$$\alpha_- = (1/4 - 4e)(1/8 - 2e), \quad \beta_+ = 1/3072, \quad W_G = 5/\alpha_-, \quad W_A = W_G - 4, \quad U_G = U_A = \beta_+/\alpha_-. \tag{17.3}$$

These are exact rational numbers. The reference rows with gate success $1/4$ and productive acceptance α_- obey $W \ge \mathbf{1} + QW$ and $U \ge \beta + QU$. Every actual gate row has greater success and every implemented productive row has acceptance at least α_- and loss contribution at most β_+. Thus the same inequalities hold directly for the implemented raw rows: a killed-potential certificate has relative perturbation constants $\rho = \eta = \zeta = 0$ relative to these conservative rows. Summing the finite-horizon inequalities, *including* the nonnegative surviving W and U terms before taking the limit, proves

$$\mathbb{E}C \le 5/\alpha_- < 161, \qquad R \le 1/(3072\alpha_-). \tag{17.4}$$

Equivalently these bounds follow from Proposition 17.1 with the conservative acceptance and numerator. A same-report changed decision lies only in $A \setminus A_{\mathrm{in}}$, whose probability is at most $1/32 - \alpha_-$. No total-variation comparison of continuous and atomic report laws was made.

For a matching all-program lower at the same budget 161 and three state-stopped labels, Theorem 9.1 gives $\mathcal{Q}_1(161/5)$. For any output r, a sublevel $t^2 + (y - r)^2 \le z$ in $[0, 1]^2$ has area at most $\pi z/2$. Integration of the selected tail gives $\mathcal{Q}_1(b) \ge 1/(\pi b)$. When $b \ge 8/\pi$, a half-disk centered at $(1/2, 0)$ of area $1/b$ stays inside the square and attains equality. Consequently

$$\frac{5}{161\pi} = R^{\Theta}_{\mathrm{st}}(161, 3) \le R_{\mathrm{dyadic}} \le \frac{1}{3072\alpha_-}. \tag{17.5}$$

The equality is the ideal state-stopped Borel optimum; the displayed dyadic observer is a separately certified nearby finite implementation. The event-stopped optimum at two labels is instead $\mathcal{Q}_2(161/5)$, and the same finite one-output construction gives only an upper bound for it.

The three threshold integers are $6145, 10239, 2047$, requiring 45 read-only bits at 15 bits each, plus the fixed comparator program and output $1/2$. Two 15-bit digitized-coordinate registers, one fresh flip bit and fixed control scratch suffice within a productive call. The three state-stopped labels require two encoding bits; the two-label event-stopped implementation requires one encoding bit; neither a counter nor a retained random seed is present. Comparisons use $O(14)$ bit operations per productive call. If a gate call costs at most t_G time and productive preparation, digitization and comparison together cost at most $t_A(14)$, expected physical execution time is at most $161 \max(t_G, t_A(14))$. The promised digitizer and balanced physical pulse costs are not free. One subsequent physical audit makes the total expected raw-call budget at most 162.

18. Causal simulation and a joint attained-error law

Proposition 18.1 (Complete common-task composition)**.** *A simulator from a source experiment to a target experiment must specify a parameter-independent strategy lift, report transformation, internal state, clock, task readout and physical-command compatibility. Suppose it uses at most S retained states, a target observer uses M, and their entire stopped scored transcript laws differ by at most ε in total variation, with convention $\|P-Q\|_{\mathrm{TV}} = \sup_A |P(A)-Q(A)|$. For loss in $[0,L]$, the composed observer uses at most SM states and its risk differs by at most $L\varepsilon$. Its expected cost bound requires its own additive-cost or killed-potential certificate. Two compatible causal simulators have total deficiency at most the sum of their deficiencies and a product state upper bound.*

Proof. The simulator and observer retain their ordered pair of states; clocks and any continuing random seeds are part of those states. Compatibility makes the composition a legal causal procedure. For a common bounded scored loss, the layer-cake formula and the total-variation convention give the risk bound. Contractivity of total variation under a common Markov kernel and the triangle inequality give the two-simulator bound on complete stopped scored laws. Nonnegative costs must instead be summed along actual executed calls or bounded by separately certified stopped potentials. A small terminal-law discrepancy alone can coexist with arbitrarily large expected waiting. Reverse risk bounds need a reverse simulator or a separate attained lower construction. □

The state product is an implementation upper, not a claim of optimal simulator size. Root-mean-square readout errors add by the triangle inequality before squaring; deterministic errors may alternatively use the bounded-loss estimate following Proposition 17.1. Neither convention makes an allowed upper bound on calibration or simulation error an unavoidable risk lower.

Proposition 18.2 (Composition with live–terminal overlap)**.** *Suppose a target program has ℓ live roles and t terminal roles. A compatible parameter-independent simulator has at most K internal states while the target is live, including every continuing phase, seed and clock. Suppose further that upon target commitment all information needed for its final exported readout is represented by one of E terminal export classes; all other live simulator state may then be discarded. The composed program has a realization with at most ℓK live and tE terminal roles. It therefore needs at most*

$$\ell K + tE \quad \textit{state-stopped labels}, \qquad \max\{\ell K, tE\} \quad \textit{event-stopped labels}.$$

Under a complete stopped/scored-law deficiency certificate ε, its bounded-loss discrepancy is at most $L\varepsilon$. For two such simulators, (K_1,E_1) and (K_2,E_2) combine to (K_1K_2, E_1E_2), with additive complete-law deficiencies. Without a terminal sufficient-class certificate, retain the full simulator state and use the safe product construction of Proposition 18.1.

Proof. Before commitment store the pair of the target live role and simulator internal state. Compatibility supplies its legal command distribution and post-report transitions, with no free port input. On commitment encode only the target output role and the sufficient export class, whose fixed decoder is part of the program. These are ℓK live and tE terminal roles. The complete stopped transcript, including the commitment event, is preserved by Lemma 10.5; apply it to those two sets. A second composition stores the corresponding triples, giving the two products. Associativity is ordinary rebracketing of finite pairs and does not assert minimality. The complete-law risk and deficiency assertions follow exactly as in Proposition 18.1. Expected time and physical-call cost require their separate certificates at each composition. □

The compared transcript includes the mapped stopping time, the event of premature commitment, all required report exports and the audit. Stopping the source before the target is allowed only to the extent covered by that complete-law error certificate; a terminal-output marginal bound cannot justify it. A stateful translator is not a free change of notation: its continuing and terminal information appear in K, E. Thus this is a resource-certified simulation law, not an assertion of invariant scalar minimal memory under arbitrary simulators.

Theorem 18.3 (Same-task attained calibration and erasure)**.** *Use the exact balanced class $D_0 = 1$, $B = 2$, with matched effort. Add an initial paid observation of an independent fair sign E_0 erased with probability 2ε, where $0 \le \varepsilon \le 1/2$, and a final paid audit of*

$$(Z, E_0, s\delta), \qquad s \in \{-1, 1\}, \quad \delta \ge 0.$$

The fixed unknown sign s affects no pre-audit kernel. Conditional on the initial reported tag, the entire core experiment contains no additional information about E_0 on the erased branch; its physical kernels may not read that hidden sign. All output coordinates are committed before this audit. Use squared Euclidean loss on the product carrier. For $N \ge \kappa + 2$, put $b = (N-2)/\kappa$. The following bounds hold for either stopping interface $\diamond$ on its indicated sufficient state range:

$$\mathcal{Q}_M(b) + 2\varepsilon + \delta^2 \le R^{\Theta}_{\mathrm{joint},\diamond}(N, M) \le \mathcal{Q}_{m_\diamond}(b) + 2\varepsilon + \delta^2, \tag{18.1}$$

$$m_{\mathrm{st}} = \lfloor (M-1)/3 \rfloor - 2 \quad (M \ge 13), \qquad m_{\mathrm{ev}} = \lfloor M/3 \rfloor \quad (M \ge 7). \tag{18.2}$$

If (13.1) *holds, this gives in both cases*

$$R^{\Theta}_{\mathrm{joint},\diamond}(N, M) \asymp F((N-2)/\kappa, M) + \delta^2 + \varepsilon. \tag{18.3}$$

The early report channel has causal deficiency exactly *ε relative to the perfect-sign channel when its virtual sign must be delivered before any core call. The stated alphabet ranges are sufficient constructions, not minimal-state assertions for the enlarged task.*

Proof. For the upper use one initial live role and three tagged copies of the two core live roles: tags $+$, $-$ and erasure. Thus there are seven live roles. There are $3m$ terminal roles, one for each tag and each core output. Output the reported sign on known branches, zero on erasure, and zero for the inaccessible calibration sign. The balanced core upper has the same risk after every preceding history, so its contribution is $\mathcal{Q}_m(b)$. The other contributions are exactly 2ε and δ^2. Initial observation and final audit each cost one. State stopping uses $7 + 3m = 1 + 3(2 + m)$ labels, whereas event stopping uses $\max\{7, 3m\}$ by Lemma 10.5. These give (18.2). The initial live role is distinct in this construction; no argument claims that every implementation needs such a separate label.

For the lower use the fixed reference core. Expected core cost is at most $N - 2$, and its terminal projected dictionary has at most M values under either semantics. Lemma 10.2 gives $\mathcal{Q}_M(b)$. On the erased branch, induction over the causal core kernels and independent observer coins leaves E_0 fair conditional on all pre-audit information, even after stopping. Any sign estimate then has conditional squared loss at least one. Average the two indistinguishable alternatives $s = \pm 1$: for any committed last coordinate r, their average loss is $\delta^2 + r^2 \ge \delta^2$. The pre-audit law, the core lower and the erased-sign lower are identical for these alternatives. The three lower contributions therefore hold simultaneously in a single worst-case total risk, not for incompatible separately optimized tasks. On the stated ranges, $m_{\mathrm{st}} \ge M/8$ and $m_{\mathrm{ev}} \ge M/4$. Apply Proposition 13.1 to obtain (18.3).

On erasure, every immediately supplied virtual sign is independent of the true fair sign. The perfect channel has event $\{\text{virtual sign equals } E_0\}$ with probability one, while every source simulator has probability at most $1 - \varepsilon$. This gives deficiency at least ε on the complete scored transcript. Reporting the known sign when available and an independent fair sign otherwise disagrees with the perfect report with probability ε. A coupling, followed by the same causal core strategy and task audit, gives the opposite bound. The early completion requirement forbids waiting for the final audit to learn E_0. □

The joint theorem is an attained construction for one specified scored task, not a universal additive theorem for arbitrary error tolerances. Theorem 11.1 and Proposition 17.1 add their own budget correction and numerical certificates. Persistent labels, simulator states, terminal export classes, scratch, program description, pulse precision, raw acquisitions and physical time remain distinct accounts.

Appendix A. Executable predictive quotients and the retained flow mechanism

Fix a declared class $\mathcal{T}$ of executable future tests, including the legal command domains, resource records and scored readouts. Suppose a countable family (T_j) determines its conditional laws. For finite visible histories h set

$$q(h) = (\mathrm{Law}(T_j \mid h))_{j\geq 1}.$$

Take Borel conditional versions on the histories under consideration. Assume the actual image $S = q(\mathsf{H})$ is Borel and admits a Borel section, legal command domains are constant on fibers, and raw report/update laws factor jointly measurably through q. These are realization hypotheses; arbitrary measurable equivalence relations need not satisfy them.

Proposition A.1 (Minimality for declared executable tests)**.** *Under these assumptions, q realizes exactly the predictive equivalence relation for $\mathcal{T}$ and carries a jointly Borel report/update kernel. Every other Borel statistic that determines these tests and admits the corresponding Borel factorization carries q measurably. For a stopped test, equality of its law follows only with the declared almost-sure termination or approximation conditions.*

Proof. Coordinate conditional laws are Borel. Their equality is precisely predictive equivalence by determination. Compose the raw kernel with a Borel section of q and push the continued history through q. Fiber constancy makes the result independent of that section and jointly Borel in state and command. Induction proves equality for finite executable prefixes. For bounded stopped readouts under almost-sure termination, truncate at time n with a distinguished surviving symbol and let $n \to \infty$; the surviving probability vanishes. For another sufficient statistic the test coordinates are constant on its fibers and the assumed factorization supplies the Borel map to q. This is minimality for the declared tests, not an unrestricted optimal-control or infinite-horizon equivalence. □

The first and second task moments in the main theorem are functions of this predictive information when its realization hypotheses hold, but they need not separate all future tests. A known finite hidden-state or quantum model has its usual conditional-state carrier; if the tests do not determine that carrier, its predictive quotient is smaller. For an unknown fixed model, the family of modelwise posteriors is a comparison object, not a single executable observer. The upper in Theorem 9.1 stores neither the exact posterior nor that family.

For comparison with classical occupation methods, we state the inherited submeasure-preserving realization separately. It is not a hidden premise of the new balanced theorem.

Proposition A.2 (Finite occupation on a Borel predictive carrier)**.** *Let S, A, Y be standard Borel with legal Borel domains, a Borel legal default policy on all states, and transition/report kernel $K(ds', dy \mid s, a)$. Let α be a prepared probability measure. Suppose a finite nonnegative legal occupation measure $\eta(ds, da)$ and a legally supported terminal measure ζ satisfy*

$$0 \leq \zeta \leq \eta K, \qquad \eta_S = \alpha + (\eta K - \zeta)_{s'}.$$

There is a stationary randomized policy with a Borel post-report stopping rule whose occupation is at most η and whose full terminal measure is exactly ζ. It terminates almost surely with finite expected count and no greater nonnegative additive costs.

Proof. Disintegrate $\eta = \xi\pi$ and choose a clipped Borel density $h = d\zeta/d(\eta K)$, zero outside the stopping domain. The continued-state kernel is $Q(s, D) = \int \pi(da \mid s) \int \mathbf{1}_D(s')(1-h)K(ds', dy \mid s, a)$. Balance is $\xi = \alpha + \xi Q$; its total mass also implies $\zeta(1) = 1$. Monotone iteration gives the actual minimal occupation $\bar{\xi} = \sum_{j\geq 0} \alpha Q^j \leq \xi$. The residual $d = \xi - \bar{\xi}$ is finite and satisfies $d = dQ$, hence $\int (1 - Q\mathbf{1})\, dd = 0$. Only rows of zero acceptance carry residual mass. With the occupation density $\chi = d\bar{\xi}/d\xi$, the realized terminal measure is $\chi(s)\zeta$, and its missing mass is exactly $\int (1 - Q\mathbf{1})\, dd = 0$. Thus the terminal measure is unchanged. The actual occupation $\bar{\eta} = \bar{\xi}\pi$ is dominated by η, proving

the count and cost bounds. Finite expected count implies almost-sure commitment. Null-set choices can be completed by any measurable legal policy on the unreachable part; existence of such a completion is part of the declared Borel legal interface. □

This is a fixed-kernel Markovization statement adjacent to [7], [8], [9]. It neither gives a finite-state implementation of an arbitrary Borel policy nor merges separately chosen unknown-model policies into one program. The common finite-state implementation here comes instead from the raw moment balance and the counted rejection rule.

Appendix B. Scope and boundary examples

The center of the paper is a general sufficient-class acquired-geometry theorem, not a classification of all positive causal experiments. Its calibrated two-sided assertion uses an actual effort witness and bounded task-row discrepancies from a separately prepared comparison, which need not be an operating model. It does not assume that a competing program stops under the comparison dynamics. The exact specialization uses a fixed admissible witness with a common productive task law and an all-strategy effort inequality, with a matched upper effort constant for exactness, together with one common task-balancing program. It does not require a common primary row for all models or all strategies. If two histories in the same alleged driver state have success probabilities $1/4$ and $3/4$, a single exact driver row cannot represent both. The present upper still works when a proved conditional attempt bound and the task-measure balance remain valid; otherwise neither the finite-driver formula nor the new theorem is licensed.

Pointwise calibration is not necessary for these minimax laws. Static hidden signs have zero residual variance after fully revealing histories, and nearly pure quantum states can do the same. Positive current-report marks then fail on those fibers. The raw reference-plus-balance argument avoids that lower-variance premise; it does not assert that every such failed calibration can be repaired. Conversely, report-only balance and an unaffordable reference preparation are insufficient, as is a reference model whose supposedly unpredictable productive state is already leaked by a preceding report.

Action aliasing is harmless only when it preserves physical ports, costs, observable inputs and legal continuation kernels. Duplicating one physical command into two names does not increase Proposition 10.4's count. A phase-dependent dispatcher that can see an otherwise unavailable port state is an extra resource, not a relabeling. The proposition is not a universal minimal-controller characterization. Nor does a live lower exclude terminal reuse. Definition 2.1 and Lemma 10.5 keep the two semantics separate; Proposition 18.2 charges any stateful translator before permitting role identification.

The finite-state upper is an explicit recursion: at the two physical port labels, issue the declared command; at the productive call, compute the virtual mark and finite decision; retain only the next label. It has no history tape, accurate hidden posterior, free real register, free time counter or externally randomized persistent program. Exact Borel descriptions and finite bit descriptions are distinguished by Proposition 17.2. Optimal joint program/time/workspace complexity remains unproved.

The two intervention realizations score the post-command audit. The Gaussian theorems preserve the original current sign or direction and use no intervention on it. The directional theorem combines a fresh-reference variance lower with a static-direction output-covering lower; it does not add separately witnessed lower losses as if one model realized both. These are different declared tasks and are never exchanged by an untyped reduction. The reference-law geometry is derived from actual preparation and its cost is included, but its existence remains an experiment-class assumption. General learning of an unknown gate lower bound, unrestricted kernel learning, arbitrary unbounded losses, nonterminating protocols, infinite-dimensional unbounded operators, and large-deviation equivalence are not conclusions of the paper. The stopped results proved here use nonnegative costs

and explicit conditional hazards; they are not substitutions of a random time into a fixed-horizon theorem.

The retained predictable-geometry manuscript contains the earlier pointwise calibration and conditional-envelope route, including nonexact acquired-mass bounds. It remains valid under its stated assumptions and is preserved in full as an optional companion. Earlier curvature, continuation, adaptive refinement, causal morphism and independent realization papers likewise keep their own hypotheses. None is used backwards to establish the central theorem here, and no historical archive is silently recertified by the present proof.

Hypothesis and proof dependencies. The measurable and effective assertions have different hypotheses:

$$\begin{gathered}\text{actual reference + all-strategy effort + exact task balance}\\ \Downarrow \quad \text{Theorem 9.1}\\ \text{attained Borel risk profiles + explicit stopping-role widths}\\ \text{uniform conditional report/task defects}\\ \Downarrow \quad \text{Theorem 11.1}\\ \text{cost-corrected approximate Borel implementation}\\ \text{certified partitions + integration + probabilities + evaluation costs}\\ \Downarrow \quad \text{Proposition 17.2}\\ \text{finite description and separately charged execution bounds.}\end{gathered}$$

The last implication is not available from Borel measurability alone. Theorem 16.1 has its own explicit raw sensor, static-direction and radial-reference assumptions, and proves its stated two-sided bounds without exact task balancing. Standard compactness, conditional projection, optimal trimming, Gaussian tails and sphere covering are used with their stated hypotheses. The principal added claim is the mass-defective accepted comparison submeasure and its two-sided risk transfer, with an actual-occupation drift extension and two raw imperfect-reference verifications. The exact stopping-resource laws, common balance, role-aware simulation and directional sensor are retained mathematical dependencies, not re-announced as new. The drift inequality is a standard telescoping device; its use here retains both selected mass and task error. No universal classification or optimal computational synthesis is inferred from these sufficient theorems.

The calibrated implication is separate from the exact diagram above:

$$\begin{gathered}\text{full productive-row calibration along actual histories}\\ \text{+ all-strategy actual effort and defect occupation}\\ \text{+ one common approximate task-balanced program}\\ \Downarrow \quad \text{Theorem 4.1}\\ \text{two-sided mass-corrected risk and explicit stopping widths}\\ \Downarrow \quad \text{Corollary 4.3}\\ \text{matching resolution law in its quantified error regime.}\end{gathered}$$

The comparison may be a preparable ideal outside the operating family, but its full current report coordinates and moment specification must be certified. A comparison chosen retrospectively from a convenient formal parameter manifold is not sufficient. Calibration acquisitions and error confidence are not free: a finite-data inference of the all-history certificates would need an additional learning theorem and its resources. At fixed positive error no arbitrarily fine matching law is asserted. The metastable extension keeps initial bad mass and inverse exit probability; it does not replace acquired support by a stationary or normalized bad stratum. Uniform approximation of a discontinuous acceptance rule is not guaranteed by pointwise convergence; Proposition 7.2 requires its actual implementation certificate. These are mathematical boundaries, not exceptions concealed in operational documentation.

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